

MANHATTAN SCALE™

SINGLE SIGN-ON GUIDE: ACTIVE DIRECTORY FEDERATION SERVICES

SINGLE SIGN-ON GUIDE: ACTIVE DIRECTORY FEDERATION SERVICES

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Revisions

Date	Description	Applies to build
March 29, 2017	Created	14.0.1358.0
November 10, 2017	Added Client secret key and Reply URL	15.0.0168.0
January 01, 2018	Added Register SCALE as Native Application	15.0.0273.0
February 14, 2018	Added Register SCALE as Native Application with AD FS	15.0.0394.0
June 13, 2018	Split to AD FS document	15.0.0836.0
July 19, 2018	Updated Register SCALE as Native Application	15.0.0836.0
July 31, 2018	Added URL load balancer	15.0.0836.0
November 6, 2018	Added Native application with scope	16.0.0100.0
May 30, 2019	Added AD FS Windows 2019 support	16.0.0715.0
Oct 09, 2020	Corrected power shell instruction for Register SCALE as Native Application	18.0.0188.0
August 2, 2022	Removed Step 2 under Configure SCALE to use ADFS as its obsolete.	

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Overview

The SCALE application requires single sign-on (SSO) to allow users access to SCALE with their standard company credentials. SCALE accomplishes this using the OAuth 2.0 protocol with OpenID Connect. SCALE supports the following SSO services:

- Microsoft Entra ID (Azure AD)
- Active Directory Federation Services (AD FS)

This guide is for setting up Single Sign on with AD FS.

Active Directory Federation Services

Active Directory Federation Services (AD FS) is a Microsoft Windows Server-based directory and identity management service.

Prerequisites

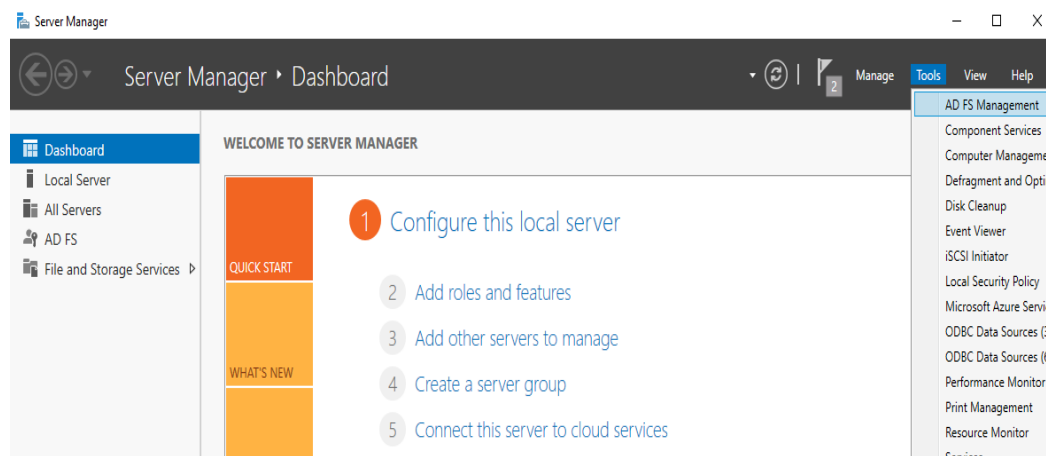
- Configured AD FS in Windows Server 2016 or 2019. (see <https://docs.microsoft.com/en-us/windows-server/identity/ad-fs/ad-fs-deployment> for more info)
- Active directory security group “SCALEUSER” with users that need access to the SCALE application or configuration web application.
- Active directory security group “SCALECONFIGUSER” with users that need access to the SCALE configuration application.
- Active directory security group “TPMUSER” with users that need access to the SCALE TPM application.

Register SCALE as an Application

For SCALE to work with AD FS, we need to register SCALE as an application in AD FS.

1. Log on to AD FS server and navigate to **Server Manager->Tools->AD FS Management**.

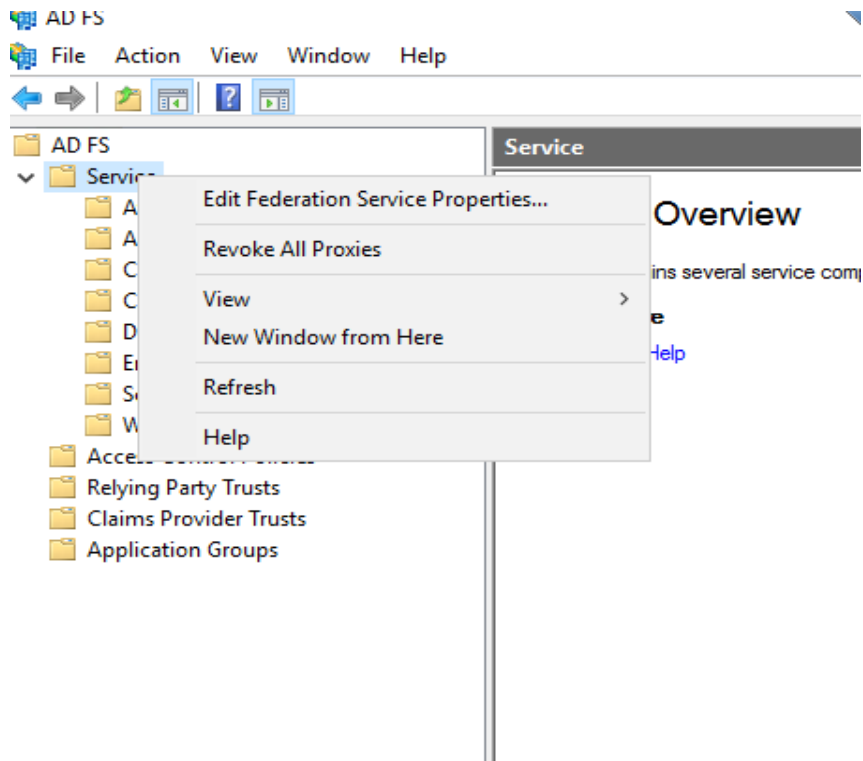
Figure 1



2. Right-click on the **Service** node and then click **Edit Federation Service Properties**.

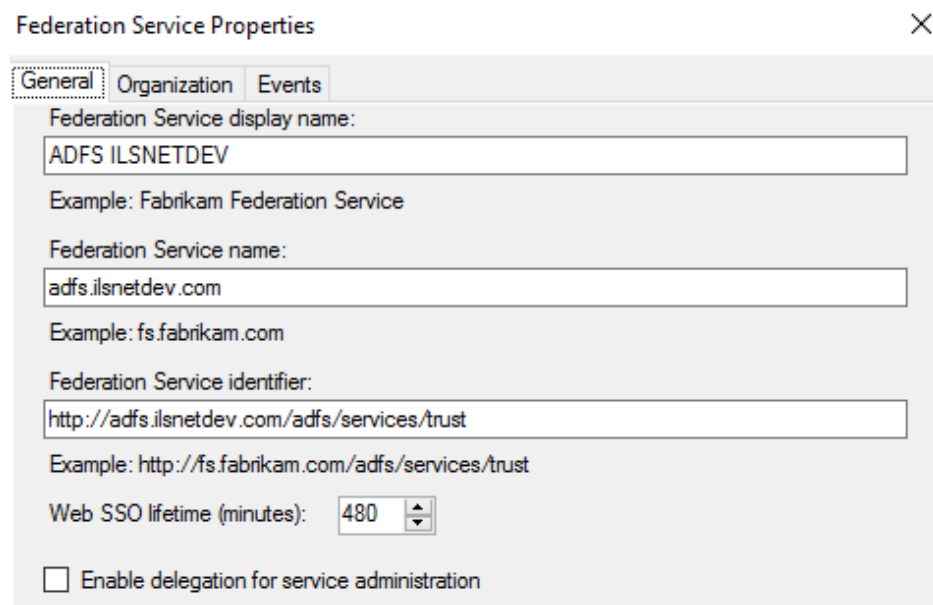
Figure 2

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3. Note the **Federation Service Name**. You will configure your application to use this later.

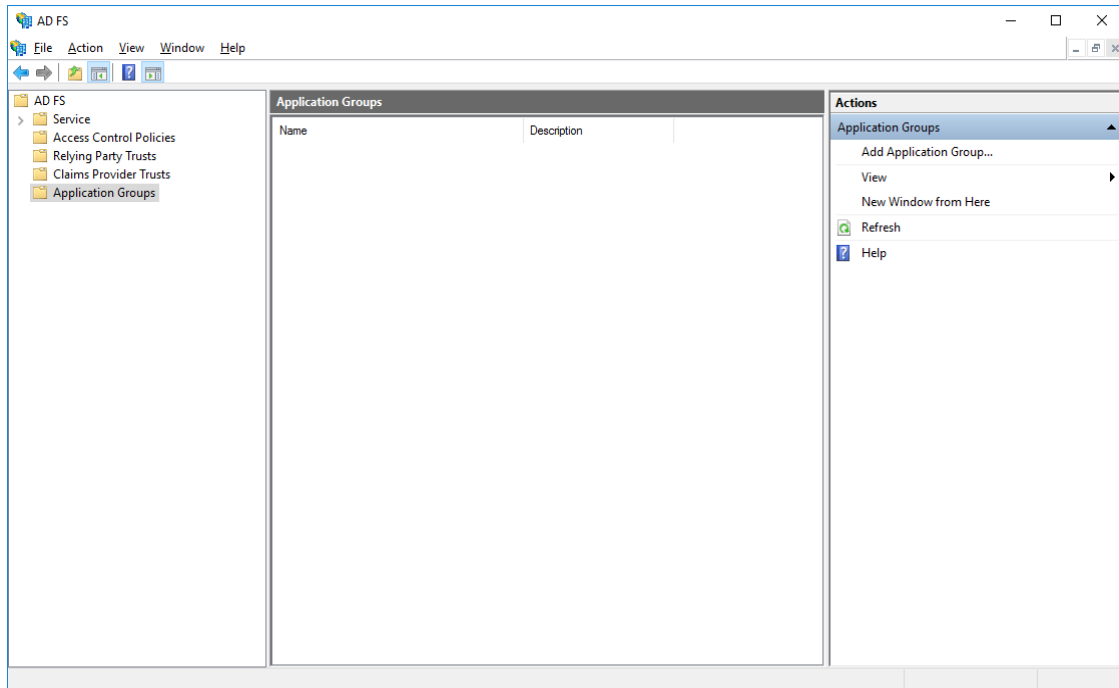
Figure 3



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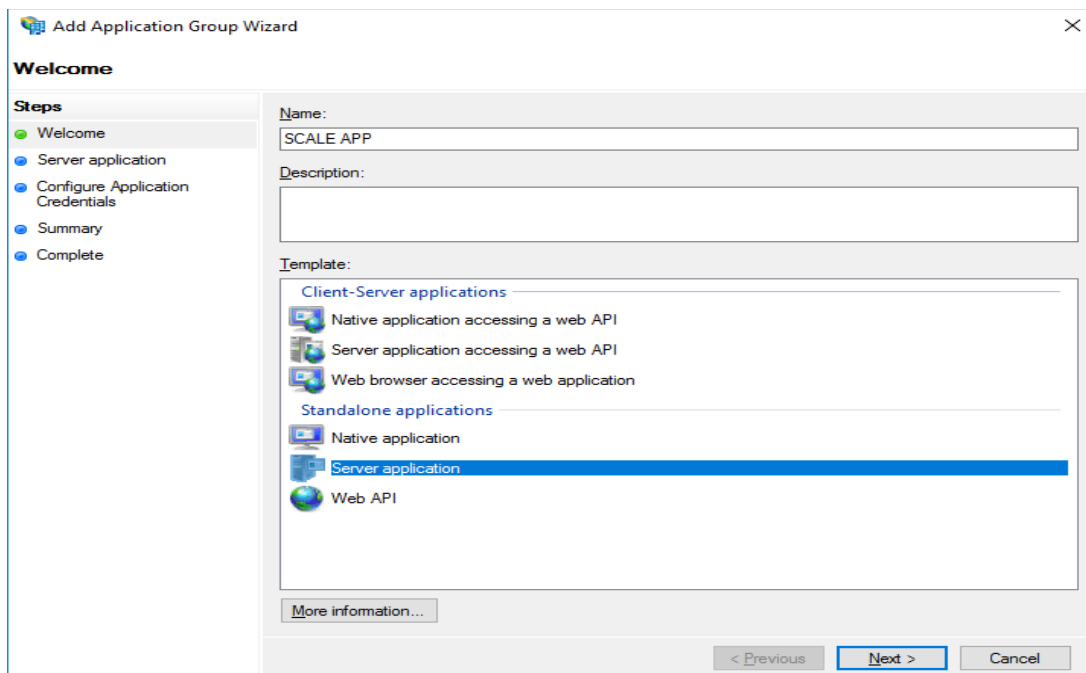
4. Select **Application Groups**. Click **Add Application Group** in the Actions pane to create a new app registration.

Figure 4



5. In the **Add Application Group Wizard** screen, enter a **Name** for the application and select the **Server Application** type in the template section. Click **Next**.

Figure 5



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6. **This step is only for customers without any SCALE environments configured.** Enter the 6 redirect URLs that were configured in lower case. Click **Add**.

Note: Lower case must be used for redirect URLs in AD FS as they are case sensitive. hostname is the fully qualified domain name pointing to SCALE server/s. To access SCALE endpoints (RF, SCALE, Web Service, TPM etc) use this hostname. In the below Example H-dalal-16.ilsnetdev.pri is the hostname pointing to the SCALE server. To access SCALE/RF/TPM/Web Services use URL `https://<hostname>/Scale/` or `https://<hostname>/rf/logon.aspx` or `https://<hostname>/TPM`. Configure the same hostname as URL load balancer in SCALE toolbox as discussed [below](#)

- `https://<hostname>/scale/security/login`
- `https://<hostname>/tpm/security/login`
- `https://<hostname>/rf/logon.aspx`
- `https://<hostname>/scale/security/postredirecturi`
- `https://<hostname>/tpm/security/postredirecturi`
- `https://<hostname>/scale/security/warehousemobilepostredirecturi/`

Figure 6

The screenshot shows the 'Add Application Group Wizard' window, specifically the 'Server application' step. The wizard has a sidebar with steps: Welcome, Server application, Configure Application Credentials, Summary, and Complete. The 'Server application' step is selected. The main area contains the following fields:

- Name:** SCALE APP - Server application
- Client Identifier:** Client identifier goes here
- Redirect URI:** A list of URLs is entered: `https://h-dalal-16.ilsnetdev.pri/scale/security/login`, `https://h-dalal-16.ilsnetdev.pri/tpm/security/login`, `https://h-dalal-16.ilsnetdev.pri/rf/logon.aspx`, and `https://h-dalal-16.ilsnetdev.pri/scale/security/postredirecturi`. The last URL is highlighted. There are 'Add' and 'Remove' buttons next to the list.
- Description:** (Empty text box)

At the bottom, there are buttons for '< Previous', 'Next >', and 'Cancel'. The 'Next >' button is highlighted.

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7. **This step is only for customers with 1 or more SCALE environments configured.** Configure the 6 redirect URLs per environment as shown below and click **Add**.

Note: Lower case must be used for redirect URLs in AD FS as they are case sensitive. hostname is the fully qualified domain name pointing to SCALE server/s. Subdomain of the hostname should be the name of the SCALE environment. To access SCALE endpoints (RF, SCALE, Web Service, TPM etc) use this hostname. In the below Example environment1.ilsnetdev.pri is the hostname pointing to the SCALE server with environment1 configured as SCALE environment. To access SCALE/RF/ TPM/Web Services use URL <https://<hostname>/Scale> or <https://<hostname>/rf/logon.aspx> or <https://<hostname>/TPM>. Configure the same hostname as URL load balancer in the SCALE toolbox as discussed below.

- <https://<hostname>/scale/security/login>
- <https://<hostname>/rf/logon.aspx>
- <https://<hostname>/scale/security/postredirecturi>
- <https://<hostname>/tpm/security/postredirecturi>
- <https://<hostname>/tpm/security/login>
- <https://<hostname>/scale/security/warehousemobilepostredirecturi/>

8. Click **Next**.

Figure 7

The screenshot shows the 'Add Application Group Wizard' window, specifically the 'Server application' step. The wizard has a sidebar with steps: Welcome, Server application (current), Configure Application Credentials, Summary, and Complete. The main area contains the following fields:

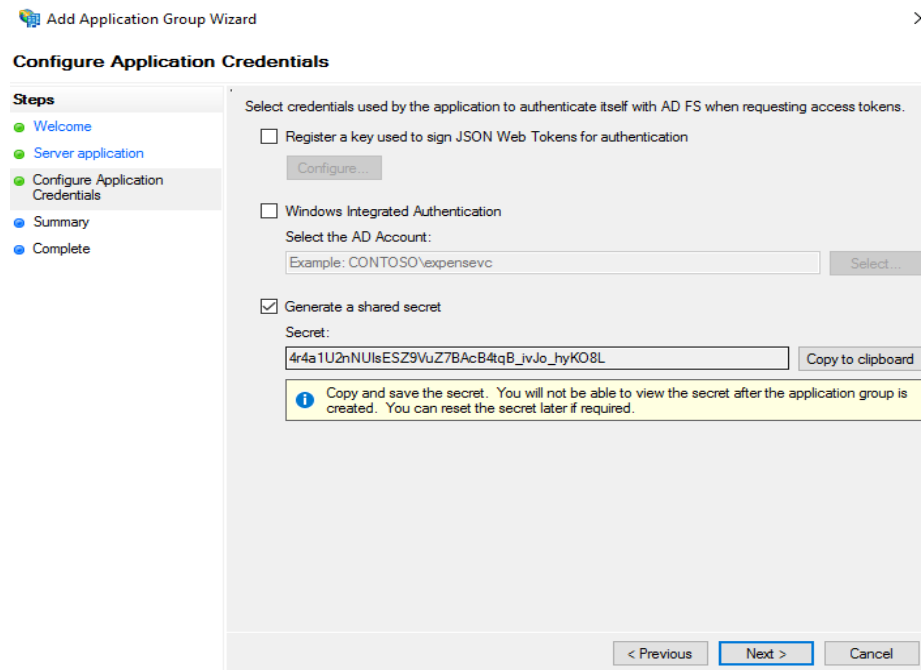
- Name:** SCALE APP - Server application
- Client Identifier:** Client identifier goes here
- Redirect URI:** A list of URLs is entered: <https://h-dalal-16.ilsnetdev.pri/scale/security/login>, <https://h-dalal-16.ilsnetdev.pri/tpm/security/login>, <https://h-dalal-16.ilsnetdev.pri/rf/logon.aspx>, and <https://h-dalal-16.ilsnetdev.pri/scale/security/postredirecturi>. There are 'Add' and 'Remove' buttons next to the list.
- Description:** (Empty text box)

At the bottom, there are buttons for '< Previous', 'Next >' (highlighted), and 'Cancel'.

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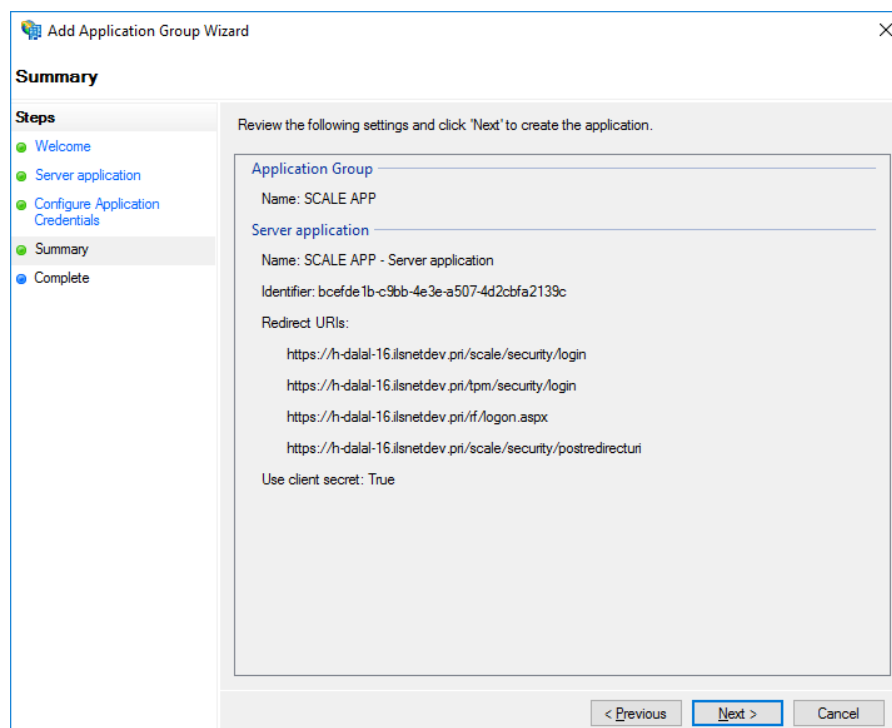
9. Select **Generate a shared secret**. Click **Copy to clipboard** and save the client secret. Click **Next**.

Figure 8



10. Note the **Identifier** for the app. You will configure your application to use this later. Click **Next**.

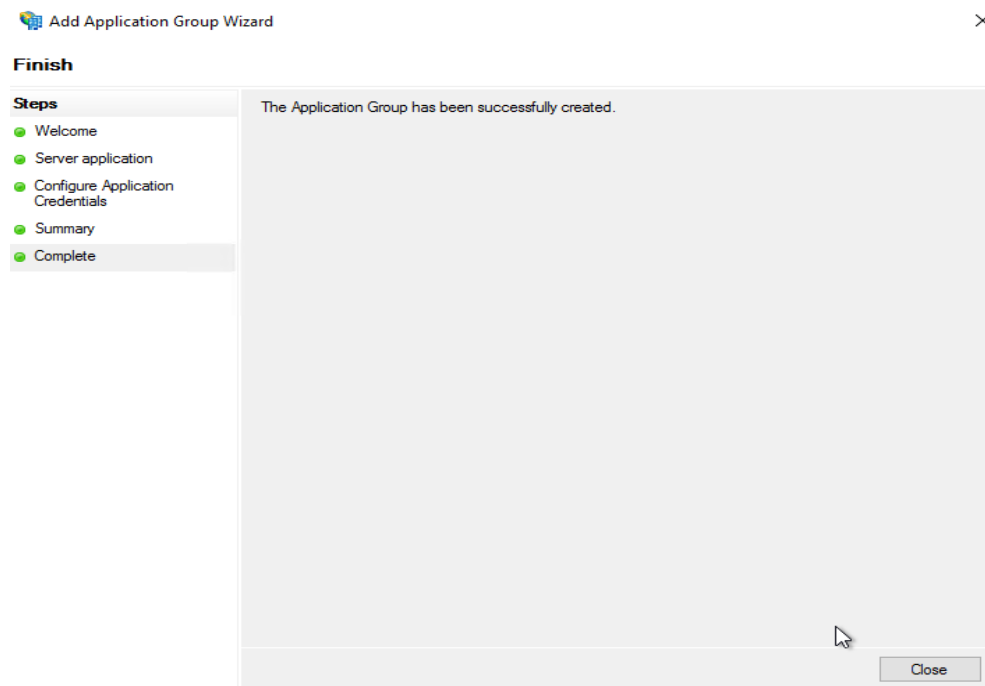
Figure 9



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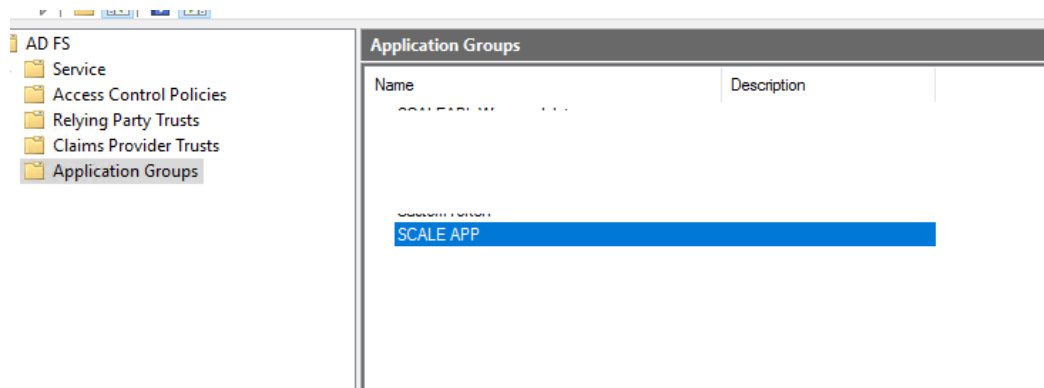
11. Click **Close**.

Figure 10



12. From the **Application Groups** window, double-click the application created above.

Figure 11



13. Click **Add Application**.

Figure 12

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The screenshot shows the 'SCALE SERVER APP Properties' dialog box with the 'General' tab selected. The 'Name' field contains 'SCALE SERVER APP'. The 'Description' field is empty. Below, the 'Applications' section contains a table with one entry: 'Server application' with a description of 'SCALE SERVER APP - Server application'. At the bottom, there are buttons for 'Add application...', 'Edit...', 'Remove', 'OK', 'Cancel', and 'Apply'.

Name	Description
Server application	SCALE SERVER APP - Server application

14. Select Web API from standalone applications and click **Next**.

Figure 13

The screenshot shows the 'Add a new application to SCALE SERVER APP' wizard. On the left, a 'Steps' list includes 'Welcome', 'Configure Web API', 'Apply Access Control Policy', 'Configure Application Permissions', 'Summary', and 'Complete'. The 'Welcome' step is currently active. The main area shows the 'Name' field with 'SCALE SERVER APP' and an empty 'Description' field. Under the 'Template' section, 'Standalone applications' are listed: 'Native application', 'Server application', and 'Web API'. 'Web API' is selected and highlighted in blue. At the bottom, there is a 'More information...' button and navigation buttons: '< Previous', 'Next >', and 'Cancel'.

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15. Paste the application identifier [copied previously](#) to the **Identifier** textbox and click **Add**.

Figure 14

The screenshot shows a dialog box titled "Add a new application to SCALE APP". The "Configure Web API" step is selected in the "Steps" list on the left. The "Name" field contains "SCALE APP - Web API". The "Identifier" field contains "bcefd61b-c9bb-4e3e-a507-4d2cbfa2139c", which is highlighted in blue. The "Add" button is visible next to the "Identifier" field. The "Description" field is empty. The "Next >" button is highlighted at the bottom right.

16. Click **Next**.

Figure 15

The screenshot shows a dialog box titled "Add a new application to SCALE SERVER APP". The "Configure Web API" step is selected in the "Steps" list on the left. The "Name" field contains "SCALE SERVER APP - Web API". The "Identifier" field contains "44267e02fac0-4e19-aa44-403ec4e9b2b2", which is highlighted in blue. The "Add" button is visible next to the "Identifier" field. The "Description" field is empty. The "Next >" button is highlighted at the bottom right.

17. Click **Next**.

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Figure 16

Add a new application to SCALE SERVER APP

Choose Access Control Policy

Steps

- Welcome
- Configure Web API
- Apply Access Control Policy
- Configure Application Permissions
- Summary
- Complete

Choose an access control policy:

Name	Description
Permit everyone	Grant access to everyone.
Permit everyone and require MFA	Grant access to everyone and require MFA f...
Permit everyone and require MFA for specific group	Grant access to everyone and require MFA f...
Permit everyone and require MFA from extranet access	Grant access to the intranet users and requir...
Permit everyone and require MFA from unauthenticated ...	Grant access to everyone and require MFA f...
Permit everyone and require MFA, allow automatic devi...	Grant access to everyone and require MFA f...
Permit everyone for intranet access	Grant access to the intranet users.
Permit specific group	Grant access to users of one or more specifi...

Policy

Permit everyone

☐ I do not want to configure the access control policy at this time. No users will be permitted access for this application.

< Previous Next > Cancel

18. Under Permitted scopes, select **allatclaims**, and **openid** and click **Next**.

Figure 16

Add a new application to SCALE APP

Configure Application Permissions

Steps

- Welcome
- Configure Web API
- Apply Access Control Policy
- Configure Application Permissions
- Summary
- Complete

Configure permissions to enable client applications to access this Web API.

Client application (caller):

Name	Description
SCALE APP - Server application	

Add... Remove

Permitted scopes:

Scope Name	Description
☒ allatclaims	Requests the access token claims in the identity token.
☐ aza	Scope allows broker client to request primary refresh token.
☐ email	Request the email claim for the signed in user.
☐ logon_cert	The logon_cert scope allows an application to request logo...
☒ openid	Request use of the OpenID Connect authorization protocol.
☐ profile	Request profile related claims for the signed in user.
☐ user_imperso...	Request permission for the application to access the resour...
☐ von_cert	The von_cert scope allows an application to request VPN ...

New scope...

< Previous Next > Cancel

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19. Click **Next**.

Figure 17

The screenshot shows a wizard window titled "Add a new application to SCALE APP". On the left, a "Steps" pane lists: Welcome, Configure Web API, Apply Access Control Policy, Configure Application Permissions, Summary (highlighted), and Complete. The main area is titled "Summary" and contains the text: "Review the following settings and click 'Next' to create the application." Below this, the settings are displayed in a structured format:

- Application Group**
 - Name: SCALE APP
- Web API**
 - Name: SCALE APP - Web API
 - Identifiers: bcefd1b-c9bb-4e3e-a507-4d2cbfa2139c
 - Access control policy: Permit everyone
 - Application permissions:
 - SCALE APP - Server application - openid allatclaims

At the bottom right, there are three buttons: "< Previous", "Next >" (highlighted with a blue border), and "Cancel".

20. Click **Close**.

Figure 18

The screenshot shows a wizard window titled "Add a new application to SCALE SERVER APP". On the left, a "Steps" pane lists: Welcome, Configure Web API, Apply Access Control Policy, Configure Application Permissions, Summary, and Complete (highlighted). The main area is titled "Finish" and contains the text: "The Application Group has been successfully created." At the bottom right, there is a "Close" button.

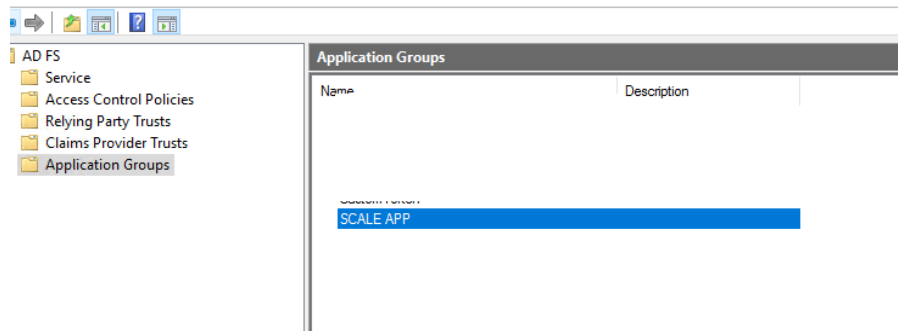
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Configure Custom Claims

We need to configure custom claims for the server/web API application to get role claims with tokens for the application.

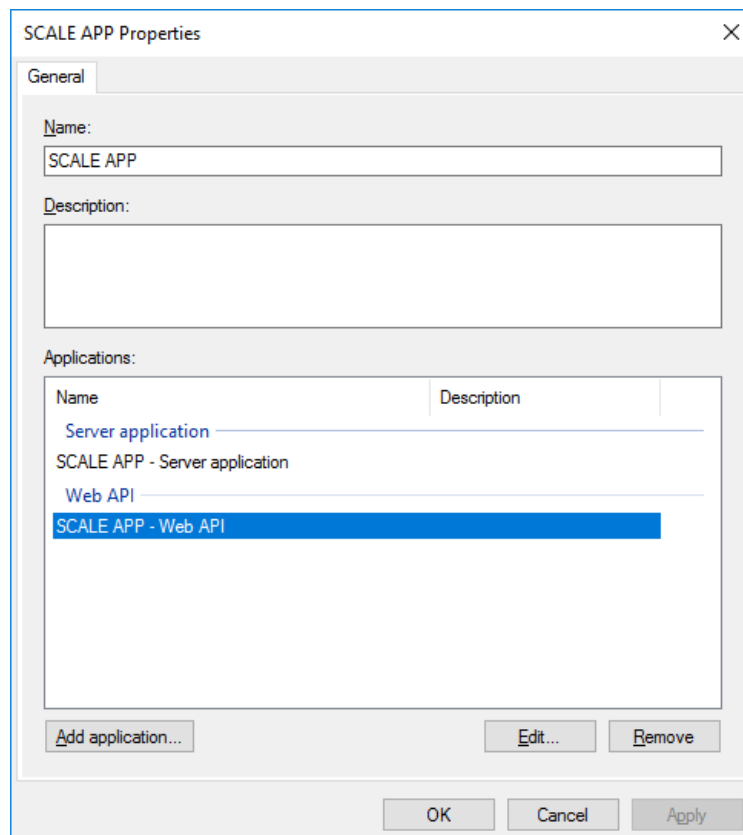
1. From AD FS Explorer, double-click the application we created above.

Figure 19



2. Double click the Web API application.

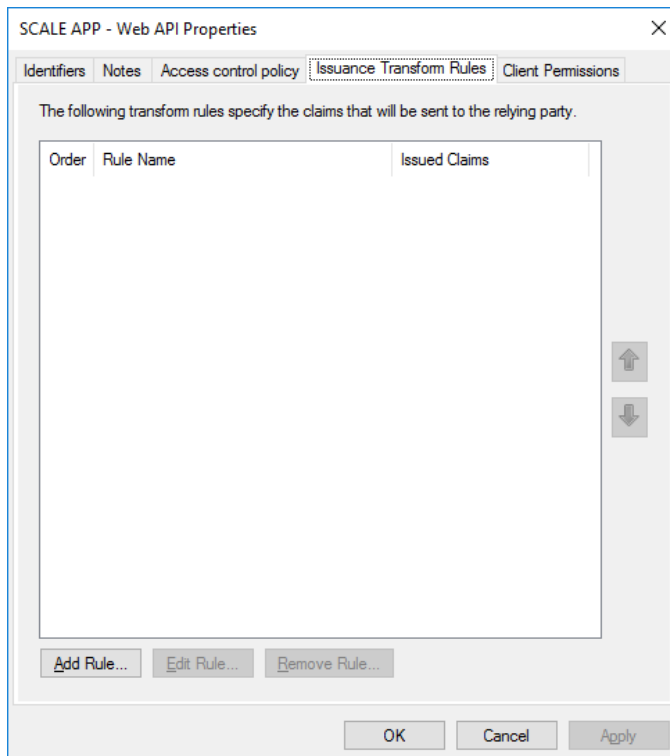
Figure 20



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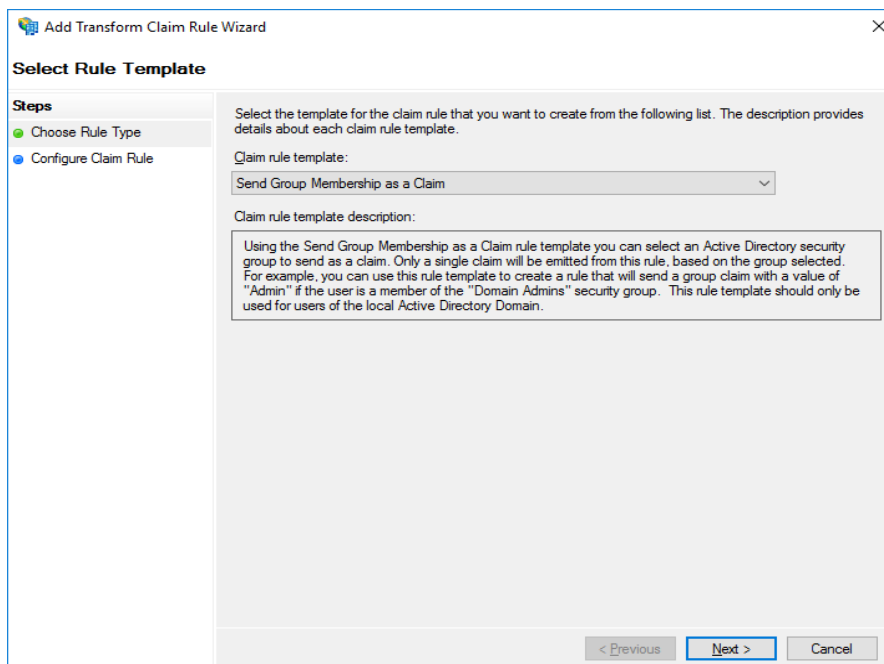
3. On the **Issuance Transform Rules** tab, click **Add Rule**

Figure 21



4. Select **Send Group Membership as a Claim** for the **Claim rule template**.

Figure 21



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5. Click **Next**.

Figure 22

The screenshot shows the 'Add Transform Claim Rule Wizard' dialog box, specifically the 'Configure Rule' step. The 'Steps' pane on the left shows 'Choose Rule Type' and 'Configure Claim Rule' as completed steps. The main area contains the following fields and options:

- Claim rule name:** An empty text box.
- Rule template:** Set to 'Send Group Membership as a Claim'.
- User's group:** An empty text box with a 'Browse...' button to its right.
- Outgoing claim type:** A dropdown menu currently showing 'Specify Claim Type...'.
- Outgoing name ID format:** A dropdown menu currently showing 'urn:ietf:params:oauth:claim-name-id-format'.
- Outgoing claim value:** An empty text box.

At the bottom right, there are three buttons: '< Previous', 'Finish' (highlighted in blue), and 'Cancel'.

6. Type the **Claim rule name** as “SCALEUSER”. Select the domain security group named “SCALEUSER”, **Outgoing claim type** as “Role” and **Outgoing claim value** as “SCALEUSER”.

Figure 23

This screenshot shows the same 'Add Transform Claim Rule Wizard' dialog box, but with the fields populated according to the instructions:

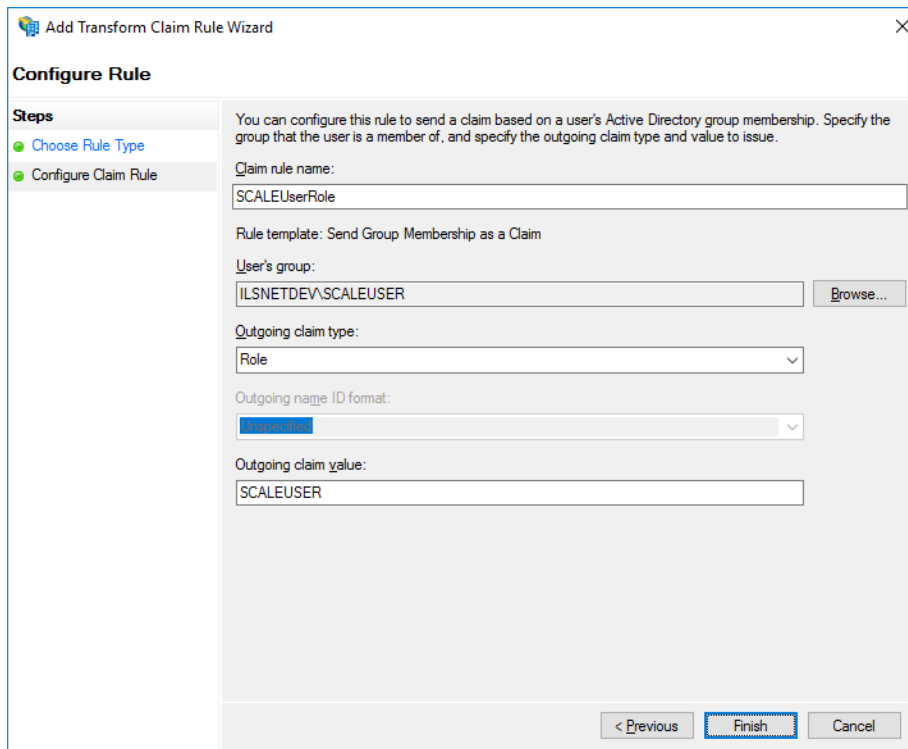
- Claim rule name:** 'SCALEUserRole'.
- User's group:** 'ILSNETDEV\SCALEUSER'.
- Outgoing claim type:** 'Role'.
- Outgoing name ID format:** 'urn:ietf:params:oauth:claim-name-id-format'.
- Outgoing claim value:** 'SCALEUSER'.

The 'Finish' button remains highlighted in blue at the bottom right.

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- Click **Finish**.

Figure 24



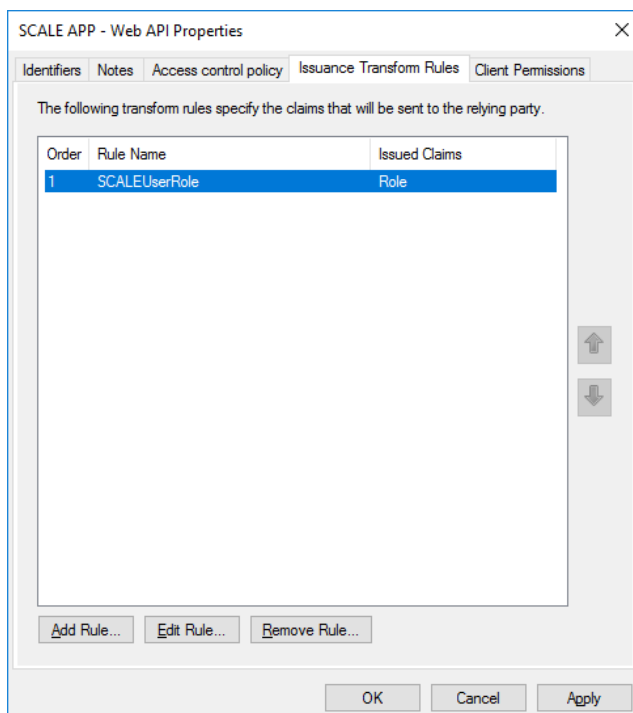
The 'Add Transform Claim Rule Wizard' window is shown at the 'Configure Rule' step. The 'Steps' pane on the left shows 'Choose Rule Type' and 'Configure Claim Rule' as completed steps. The main area contains the following fields and options:

- Claim rule name:** SCALEUserRole
- Rule template:** Send Group Membership as a Claim
- User's group:** ILSNETDEV\SCALEUSER (with a 'Browse...' button)
- Outgoing claim type:** Role (selected in a dropdown)
- Outgoing name ID format:** urn:ietf:params:oauth:claim-name-id-format (selected in a dropdown)
- Outgoing claim value:** SCALEUSER

At the bottom, there are buttons for '< Previous', 'Finish' (highlighted with a red border), and 'Cancel'.

- Click **Add Rule** on the **Issuance Transform Rules** tab.

Figure 25



The 'SCALE APP - Web API Properties' window is shown with the 'Issuance Transform Rules' tab selected. The tab shows a list of transform rules:

Order	Rule Name	Issued Claims
1	SCALEUserRole	Role

Below the table are buttons for 'Add Rule...', 'Edit Rule...', and 'Remove Rule...'. At the bottom of the window are 'OK', 'Cancel', and 'Apply' buttons.

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9. Select **Send Group Membership as a Claim** for the **Claim** rule template.

Figure 26

The screenshot shows the 'Add Transform Claim Rule Wizard' dialog box. The title bar reads 'Add Transform Claim Rule Wizard'. The 'Steps' pane on the left has two items: 'Choose Rule Type' (selected with a green dot) and 'Configure Claim Rule' (with a blue dot). The main area is titled 'Select Rule Template'. It contains the text: 'Select the template for the claim rule that you want to create from the following list. The description provides details about each claim rule template.' Below this is a 'Claim rule template:' dropdown menu with 'Send Group Membership as a Claim' selected. Underneath is a 'Claim rule template description:' box containing the text: 'Using the Send Group Membership as a Claim rule template you can select an Active Directory security group to send as a claim. Only a single claim will be emitted from this rule, based on the group selected. For example, you can use this rule template to create a rule that will send a group claim with a value of "Admin" if the user is a member of the "Domain Admins" security group. This rule template should only be used for users of the local Active Directory Domain.' At the bottom right are three buttons: '< Previous' (disabled), 'Next >' (highlighted with a blue border), and 'Cancel' (disabled).

10. Click **Next**.

Figure 27

The screenshot shows the 'Add Transform Claim Rule Wizard' dialog box at the 'Configure Rule' step. The title bar reads 'Add Transform Claim Rule Wizard'. The 'Steps' pane on the left has three items: 'Choose Rule Type' (green dot), 'Configure Claim Rule' (blue dot), and 'Configure Rule' (green dot). The main area is titled 'Configure Rule'. It contains the text: 'You can configure this rule to send a claim based on a user's Active Directory group membership. Specify the group that the user is a member of, and specify the outgoing claim type and value to issue.' Below this are several fields: 'Claim rule name:' with an empty text box; 'Rule template: Send Group Membership as a Claim'; 'User's group:' with an empty text box and a 'Browse...' button; 'Outgoing claim type:' with a dropdown menu showing 'Specify Claim Type...'; 'Outgoing name ID format:' with a dropdown menu showing 'urn:oasis:names:tc:SAML:2.0:nameid-format:emailAddress'; and 'Outgoing claim value:' with an empty text box. At the bottom right are three buttons: '< Previous' (disabled), 'Finish' (highlighted with a blue border), and 'Cancel' (disabled).

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11. Type the **Claim rule name**. Select domain security group named as “TPMUSER”, **Outgoing claim type** as “Role” and **Outgoing claim value** as “TPMUSER”.

Figure 28

The screenshot shows the 'Add Transform Claim Rule Wizard' dialog box, specifically the 'Configure Rule' step. The 'Steps' pane on the left shows 'Choose Rule Type' and 'Configure Claim Rule' as completed steps. The main area contains the following fields and options:

- Claim rule name:** TPMUsersRole
- Rule template:** Send Group Membership as a Claim
- User's group:** ILSNETDEV\TPMUSER (with a 'Browse...' button)
- Outgoing claim type:** Role (selected from a dropdown)
- Outgoing name ID format:** Unspecified (selected from a dropdown)
- Outgoing claim value:** TPMUSER

At the bottom, there are three buttons: '< Previous', 'Finish' (highlighted with a blue border), and 'Cancel'.

12. Click **Finish**.

Figure 29

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The screenshot shows the 'Add Transform Claim Rule Wizard' window, specifically the 'Configure Rule' step. The window has a title bar with the text 'Add Transform Claim Rule Wizard' and a close button. On the left, there is a 'Steps' pane with two items: 'Choose Rule Type' (highlighted with a green dot) and 'Configure Claim Rule' (also highlighted with a green dot). The main area of the wizard contains the following fields and controls:

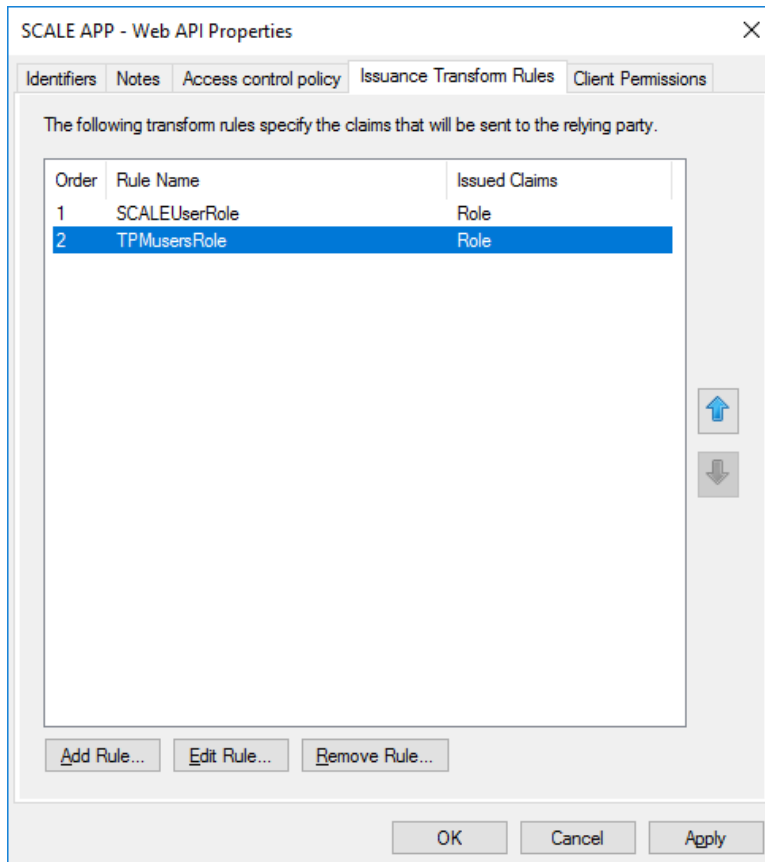
- Claim rule name:** A text box containing 'TPMusersRole'.
- Rule template:** A dropdown menu showing 'Send Group Membership as a Claim'.
- User's group:** A text box containing 'ILSNETDEV\TPMUSER' and a 'Browse...' button to its right.
- Outgoing claim type:** A dropdown menu showing 'Role'.
- Outgoing name ID format:** A dropdown menu showing 'Unspecified'.
- Outgoing claim value:** A text box containing 'TPMUSER'.

At the bottom right of the wizard, there are three buttons: '< Previous', 'Finish' (highlighted with a blue border), and 'Cancel'.

13. Click **Apply** on the **Issuance Transform Rules** tab.

Figure 30

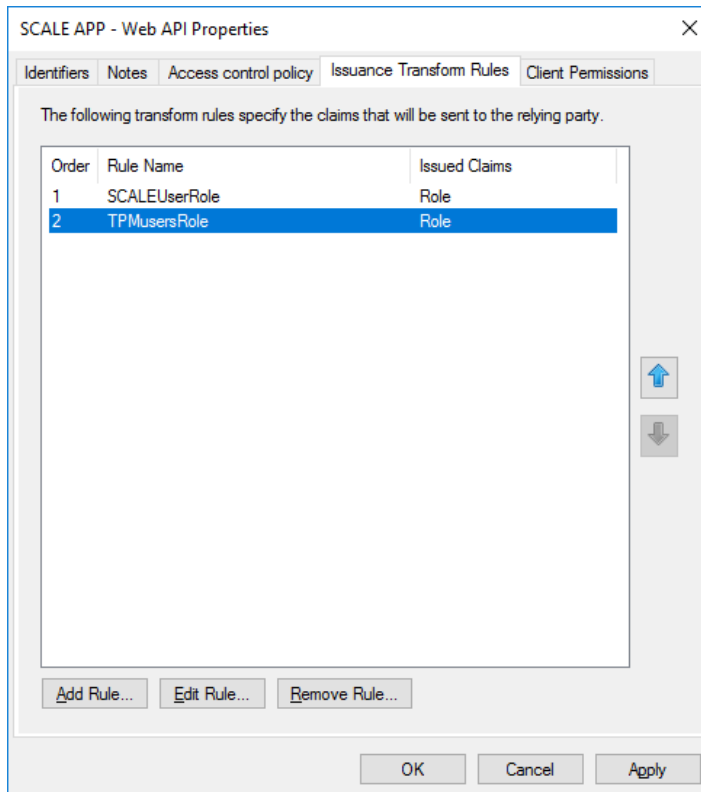
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14. Click **OK**.
15. Click **Add Rule** on the **Issuance Transform Rules** tab.

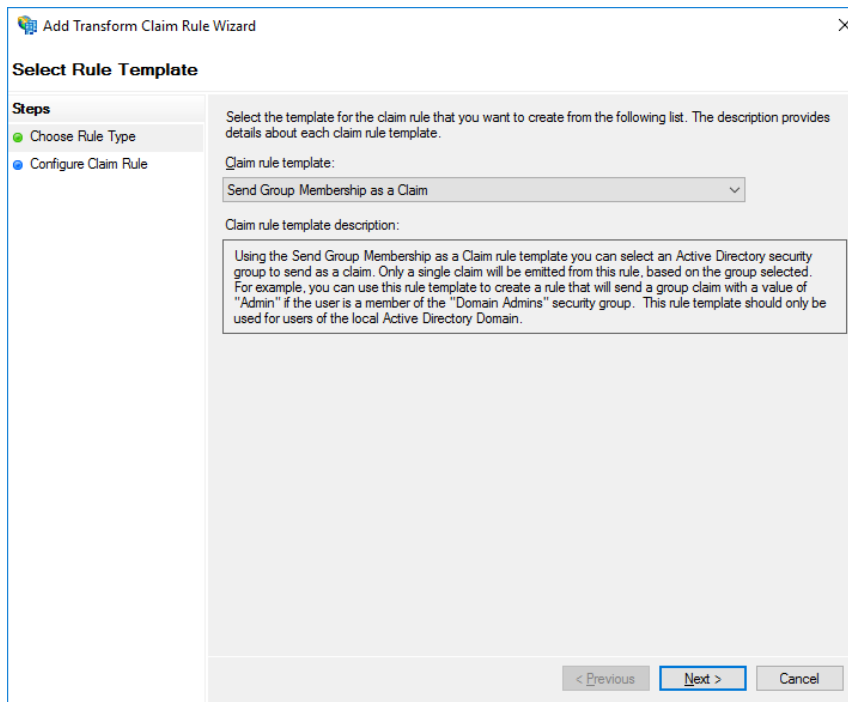
Figure 25

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16. Select **Send Group Membership as a Claim** for the **Claim rule template**.

Figure 26



17. Click **Next**.

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Figure 27

The screenshot shows the 'Add Transform Claim Rule Wizard' dialog box, specifically the 'Configure Rule' step. The 'Steps' pane on the left shows 'Choose Rule Type' and 'Configure Claim Rule' as completed steps. The main area contains the following fields and options:

- Claim rule name:** An empty text box.
- Rule template:** Set to 'Send Group Membership as a Claim'.
- User's group:** An empty text box with a 'Browse...' button to its right.
- Outgoing claim type:** A dropdown menu currently showing 'Specify Claim Type...'.
- Outgoing name ID format:** A dropdown menu currently showing 'Unspecified'.
- Outgoing claim value:** An empty text box.

At the bottom right, there are three buttons: '< Previous', 'Finish' (highlighted with a blue border), and 'Cancel'.

18. Type the **Claim rule name**. Select domain security group named as “SCALECONFIGUSER”, **Outgoing claim type** as “Role” and **Outgoing claim value** as “CONFIGUSER”.

Figure 28

This screenshot shows the same 'Add Transform Claim Rule Wizard' dialog box, but with the fields populated as per the instructions in step 18:

- Claim rule name:** 'ConfigUser'
- User's group:** 'ILSNETDEV\SCALECONFIGUSER' (with the 'Browse...' button highlighted)
- Outgoing claim type:** 'Role' (selected in the dropdown)
- Outgoing name ID format:** 'Unspecified' (selected in the dropdown)
- Outgoing claim value:** 'CONFIGUSER'

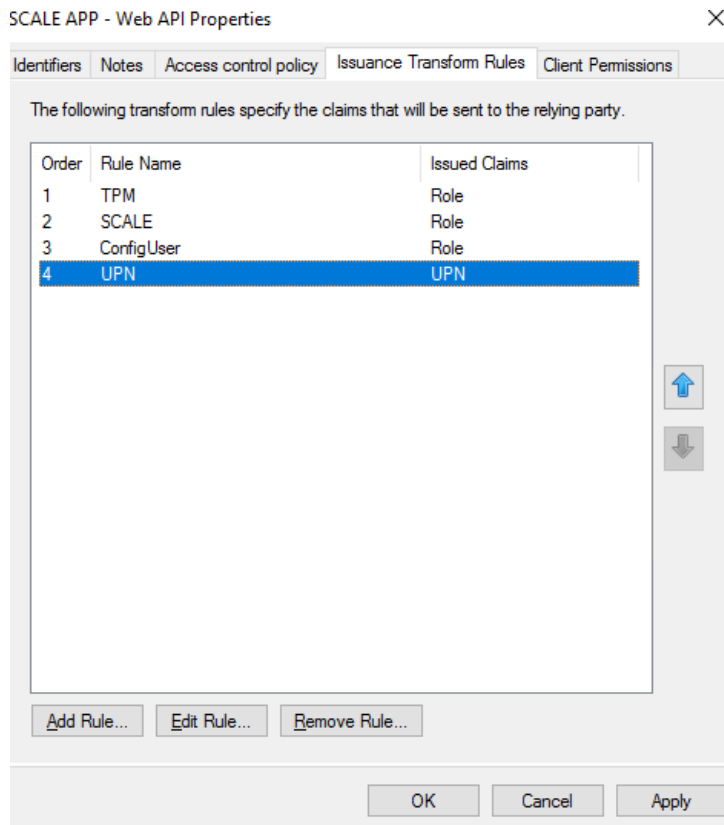
At the bottom, there are three buttons: 'View Rule Language...', 'OK', and 'Cancel'.

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19. Click **Finish**.

20. Click **Apply** on the **Issuance Transform Rules** tab.

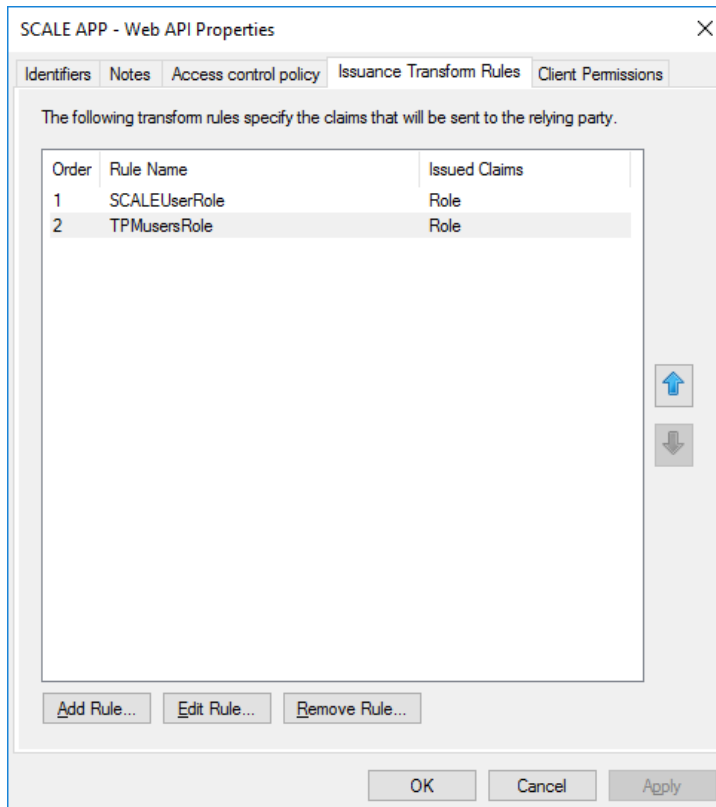
Figure 30



21. Click **OK**.

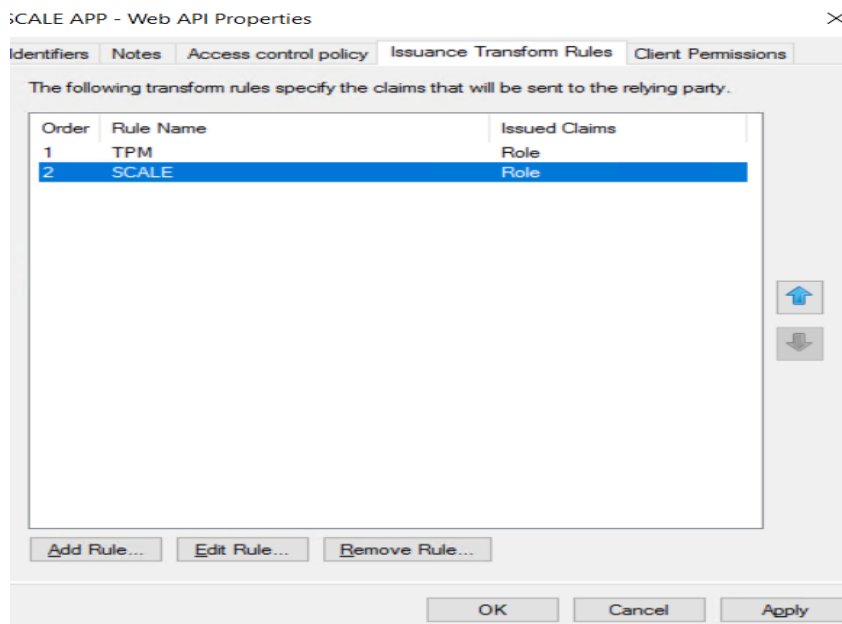
Figure 31

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22. Click **Add Rule** on the **Issuance Transform Rules** tab.

Figure 32



23. Select **Send LDAP Attributes as claim** and Click Next.

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Figure 33

The screenshot shows the 'Add Transform Claim Rule Wizard' dialog box. The title bar reads 'Add Transform Claim Rule Wizard' with a close button. The main heading is 'Select Rule Template'. On the left, a 'Steps' pane shows two steps: 'Choose Rule Type' (selected with a green dot) and 'Configure Claim Rule' (unselected with a blue dot). The main area contains instructions: 'Select the template for the claim rule that you want to create from the following list. The description provides details about each claim rule template.' Below this is a 'Claim rule template:' label and a dropdown menu showing 'Send LDAP Attributes as Claims'. A 'Claim rule template description:' label is followed by a text box containing detailed instructions about using the 'Send LDAP Attribute as Claims' rule template to extract attribute values from an LDAP attribute store like Active Directory and send them as claims to a relying party. At the bottom right, there are three buttons: '< Previous' (disabled), 'Next >' (active/highlighted), and 'Cancel'.

24. Populate the screen as below and Click **Finish**

Figure 34

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The screenshot shows the 'Add Transform Claim Rule Wizard' window, specifically the 'Configure Rule' step. The window has a title bar with a close button (X) and a subtitle 'Add Transform Claim Rule Wizard'. On the left, there is a 'Steps' pane with two items: 'Choose Rule Type' (highlighted with a green dot) and 'Configure Claim Rule' (also with a green dot). The main area contains instructions: 'You can configure this rule to send the values of LDAP attributes as claims. Select an attribute store from which to extract LDAP attributes. Specify how the attributes will map to the outgoing claim types that will be issued from the rule.' Below this, there are fields for 'Claim rule name:' (containing 'UPN'), 'Rule template:' (set to 'Send LDAP Attributes as Claims'), and 'Attribute store:' (a dropdown menu set to 'Active Directory'). A section titled 'Mapping of LDAP attributes to outgoing claim types:' contains a table with two columns: 'LDAP Attribute (Select or type to add more)' and 'Outgoing Claim Type (Select or type to add more)'. The table has two rows: the first row shows 'User-Principal-Name' mapped to 'UPN', and the second row is marked with an asterisk (*) and is currently empty. At the bottom right, there are three buttons: '< Previous', 'Finish' (highlighted with a blue border), and 'Cancel'.

Configure Rule

Steps

- Choose Rule Type
- Configure Claim Rule

You can configure this rule to send the values of LDAP attributes as claims. Select an attribute store from which to extract LDAP attributes. Specify how the attributes will map to the outgoing claim types that will be issued from the rule.

Claim rule name: UPN

Rule template: Send LDAP Attributes as Claims

Attribute store: Active Directory

Mapping of LDAP attributes to outgoing claim types:

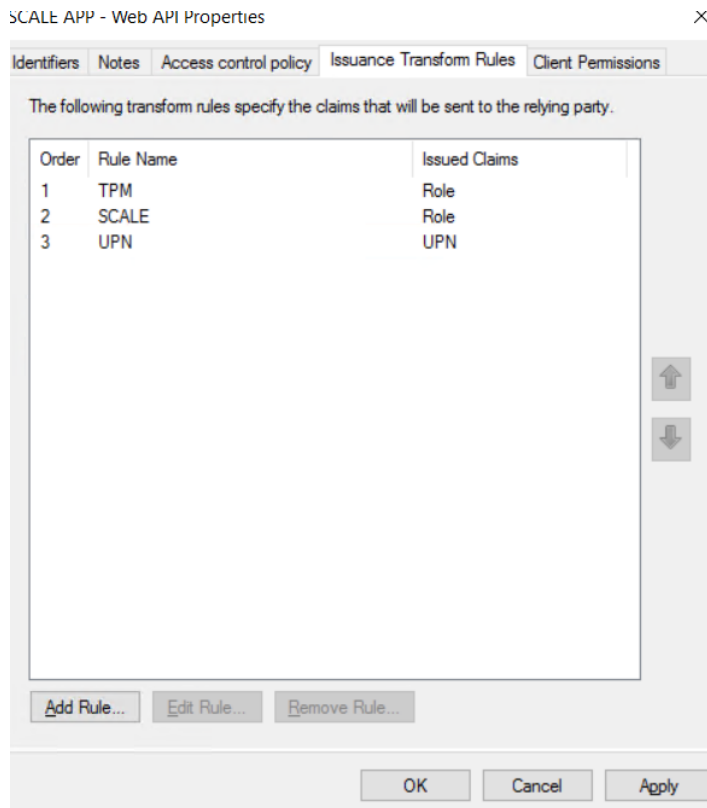
	LDAP Attribute (Select or type to add more)	Outgoing Claim Type (Select or type to add more)
▶	User-Principal-Name	UPN
*		

< Previous Finish Cancel

25. Click **Apply** and **Ok**.

Figure 35

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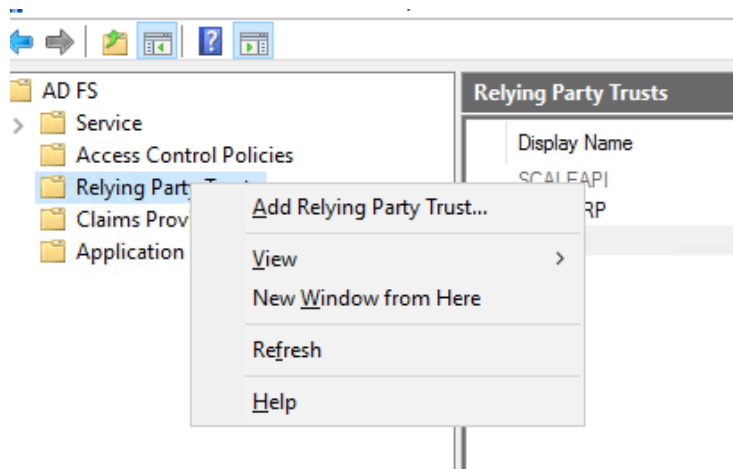


Configure Relying Party Trust

We need to configure a Relying Party Trust for server-to-server calls to get a custom claim for the application.

1. From AD FS explorer, right-click the **Relying Party Trust** folder and click **Add Relying Party Trust**.

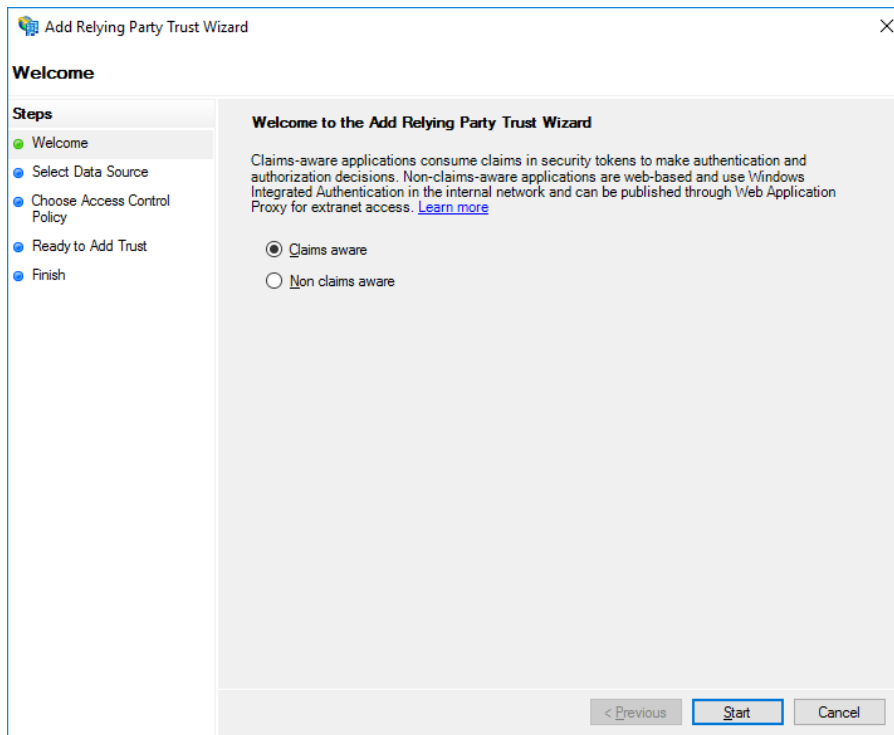
Figure 36



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2. Select **Claims aware** and click **Start**.

Figure 37



3. Select **Enter data about the relying party manually** and click **Next**.

Figure 38

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The screenshot shows the 'Add Relying Party Trust Wizard' window, specifically the 'Select Data Source' step. The 'Steps' pane on the left lists the following steps: Welcome, Select Data Source (current), Specify Display Name, Configure Certificate, Configure URL, Configure Identifiers, Choose Access Control Policy, Ready to Add Trust, and Finish. The main area contains the instruction: 'Select an option that this wizard will use to obtain data about this relying party:'. There are three radio button options: 1. 'Import data about the relying party published online or on a local network' (unselected). Description: 'Use this option to import the necessary data and certificates from a relying party organization that publishes its federation metadata online or on a local network.' Input: 'Federation metadata address (host name or URL):' with a text box and an example: 'Example: fs.contoso.com or https://www.contoso.com/app'. 2. 'Import data about the relying party from a file' (unselected). Description: 'Use this option to import the necessary data and certificates from a relying party organization that has exported its federation metadata to a file. Ensure that this file is from a trusted source. This wizard will not validate the source of the file.' Input: 'Federation metadata file location:' with a text box and a 'Browse...' button. 3. 'Enter data about the relying party manually' (selected). Description: 'Use this option to manually input the necessary data about this relying party organization.' At the bottom are '< Previous', 'Next >', and 'Cancel' buttons.

4. Enter the **Display name** and click **Next**.

Figure 39

The screenshot shows the 'Add Relying Party Trust Wizard' window, specifically the 'Specify Display Name' step. The 'Steps' pane on the left lists the following steps: Welcome, Select Data Source, Specify Display Name (current), Configure Certificate, Configure URL, Configure Identifiers, Choose Access Control Policy, Ready to Add Trust, and Finish. The main area contains the instruction: 'Enter the display name and any optional notes for this relying party.' There are two input fields: 'Display name:' with a text box containing 'SCALERPT' and 'Notes:' with a larger text box. At the bottom are '< Previous', 'Next >', and 'Cancel' buttons.

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5. Click **Next**.

Figure 40

The screenshot shows the 'Add Relying Party Trust Wizard' window, specifically the 'Configure Certificate' step. On the left, a 'Steps' pane lists the wizard's steps: Welcome, Select Data Source, Specify Display Name, Configure Certificate (highlighted), Configure URL, Configure Identifiers, Choose Access Control Policy, Ready to Add Trust, and Finish. The main area contains instructions: 'Specify an optional token encryption certificate. The token encryption certificate is used to encrypt the claims that are sent to this relying party. The relying party will use the private key of this certificate to decrypt the claims that are sent to it. To specify the certificate, click Browse.' Below this is a form with labels for 'Issuer:', 'Subject:', 'Effective date:', and 'Expiration date:'. At the bottom of this form are 'View...', 'Browse...' (highlighted), and 'Remove' buttons. At the very bottom of the wizard are '< Previous', 'Next >', and 'Cancel' buttons.

6. Click **Next**.

Figure 41

The screenshot shows the 'Add Relying Party Trust Wizard' window, specifically the 'Configure URL' step. The 'Steps' pane on the left highlights 'Configure URL'. The main area contains instructions: 'AD FS supports the WS-Trust, WS-Federation and SAML 2.0 WebSSO protocols for relying parties. If WS-Federation, SAML, or both are used by the relying party, select the check boxes for them and specify the URLs to use. Support for the WS-Trust protocol is always enabled for a relying party.' There are two sections, each with a checkbox and a text field. The first section is for 'WS-Federation Passive protocol', with an example URL 'https://fs.contoso.com/adfs/ls/'. The second section is for 'SAML 2.0 WebSSO protocol', with an example URL 'https://www.contoso.com/adfs/ls/'. At the bottom of the wizard are '< Previous', 'Next >' (highlighted), and 'Cancel' buttons.

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7. Enter **Relying party trust identifier** as “SCALERPT”. Make a note of this name as it will be required later in SCALE configuration.

Figure 43

The screenshot shows the 'Add Relying Party Trust Wizard' window, specifically the 'Configure Identifiers' step. On the left, a 'Steps' pane lists the wizard's stages: Welcome, Select Data Source, Specify Display Name, Configure Certificate, Configure URL, Configure Identifiers (currently selected), Choose Access Control Policy, Ready to Add Trust, and Finish. The main area contains instructions: 'Relying parties may be identified by one or more unique identifier strings. Specify the identifiers for this relying party trust.' Below this, the 'Relying party trust identifier:' label is followed by a text box containing 'SCALERPT' and an 'Add' button. An example URL 'https://fs.contoso.com/adfs/services/trust' is provided. A larger list box for 'Relying party trust identifiers:' is empty, with a 'Remove' button to its right. At the bottom are '< Previous', 'Next >', and 'Cancel' buttons.

8. Click **Add** and then click **Next**.

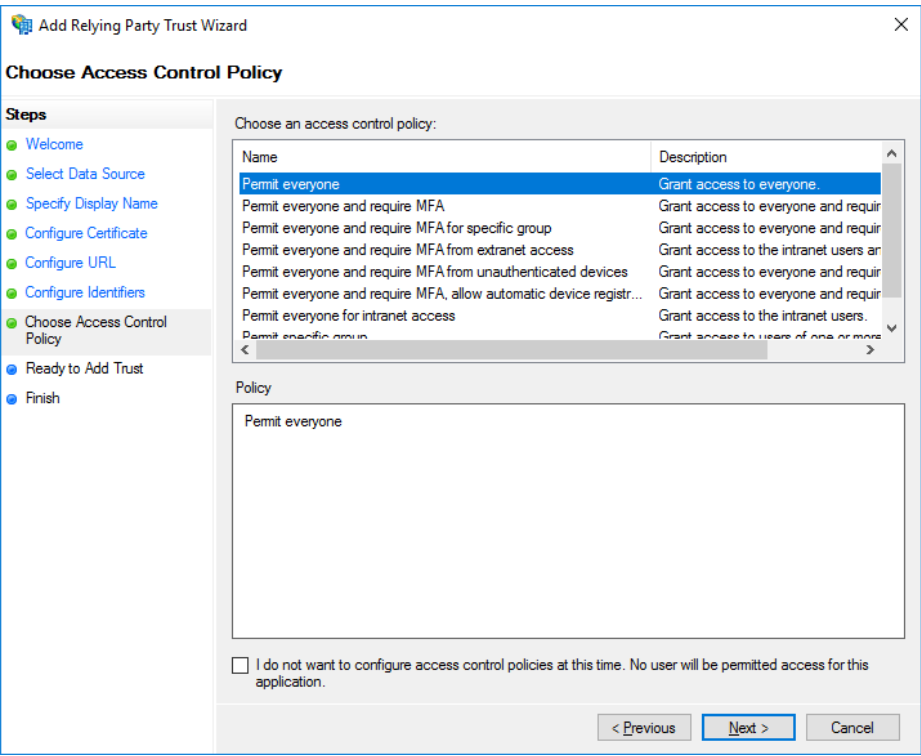
Figure 44

This screenshot shows the same 'Add Relying Party Trust Wizard' window as Figure 43, but after the 'Add' button has been clicked. The 'Relying party trust identifier' field still contains 'SCALERPT'. The 'Add' button is now disabled. The 'Relying party trust identifiers:' list box now contains 'SCALERPT', which is highlighted with a blue selection bar. The 'Remove' button is still present. The 'Next >' button at the bottom is now highlighted with a blue border, indicating it is the next step in the wizard.

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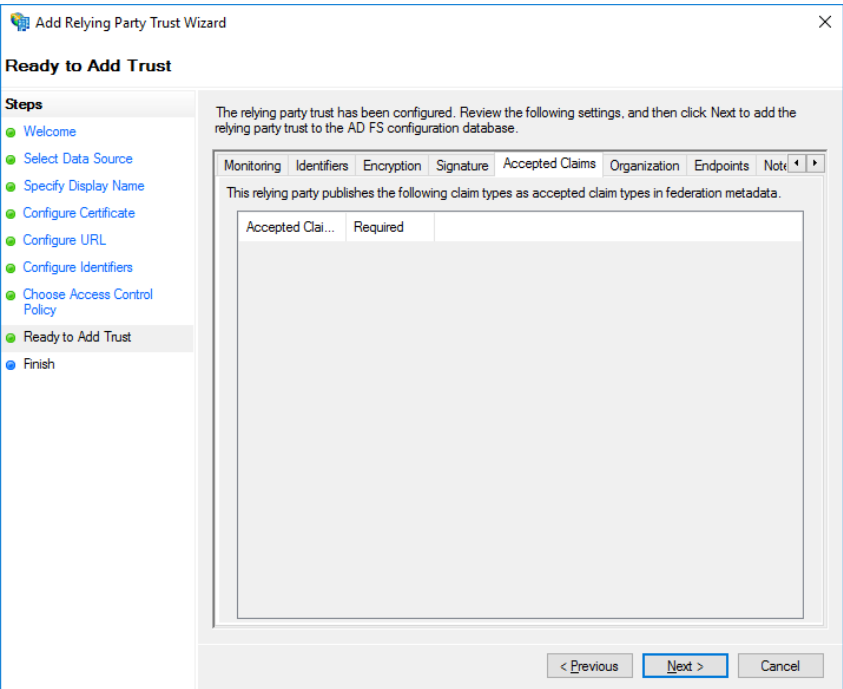
9. Click **Next**.

Figure 45



10. Click **Next**.

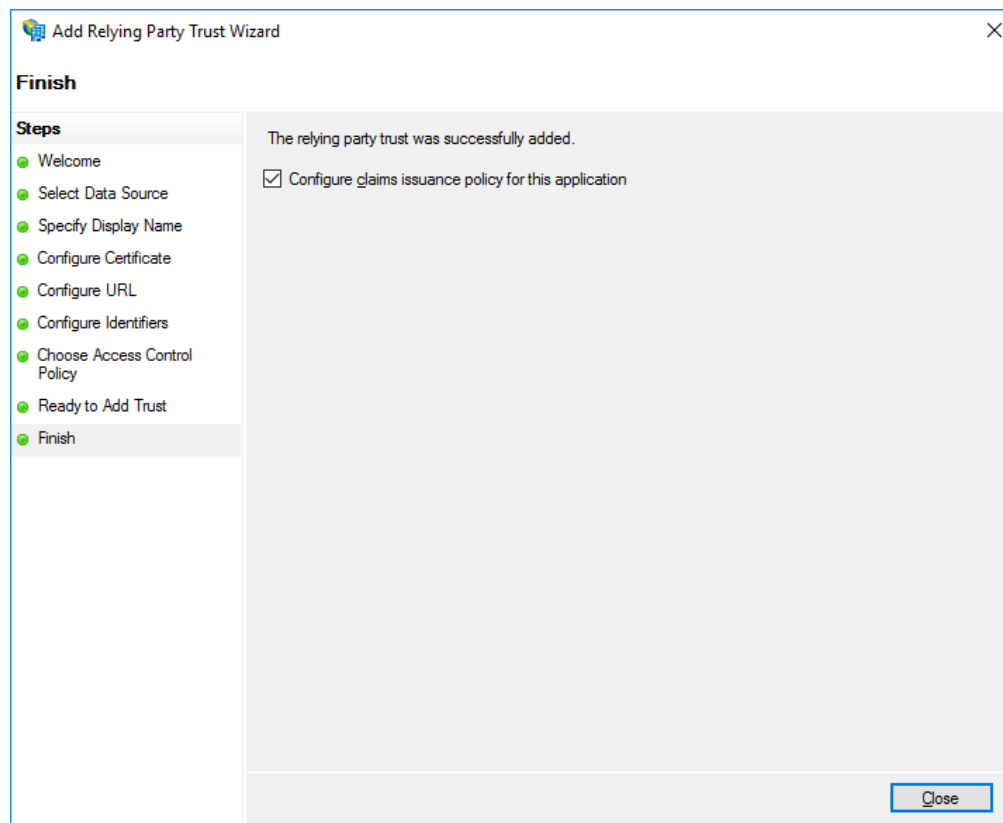
Figure 46



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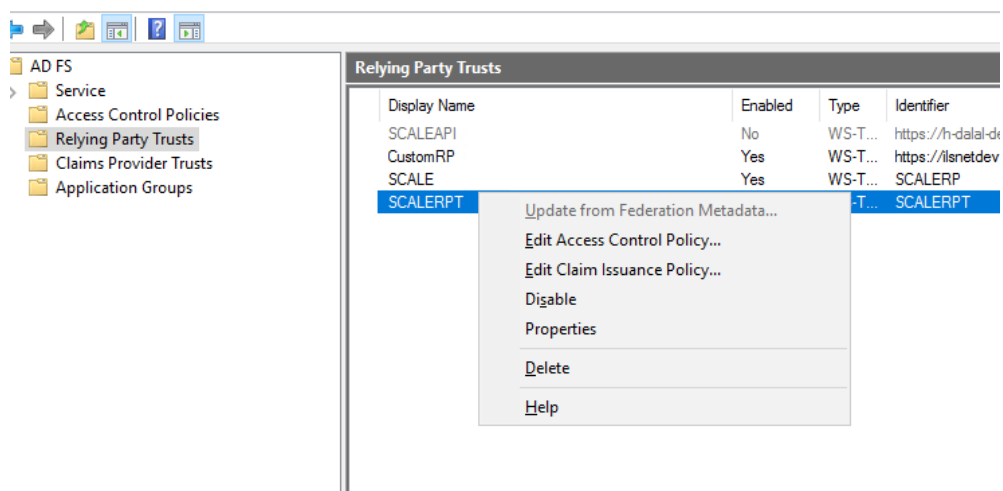
11. Click **Close**.

Figure 47



12. Right click on the newly created relying party trust and click **Edit Claim Issuance Policy**.

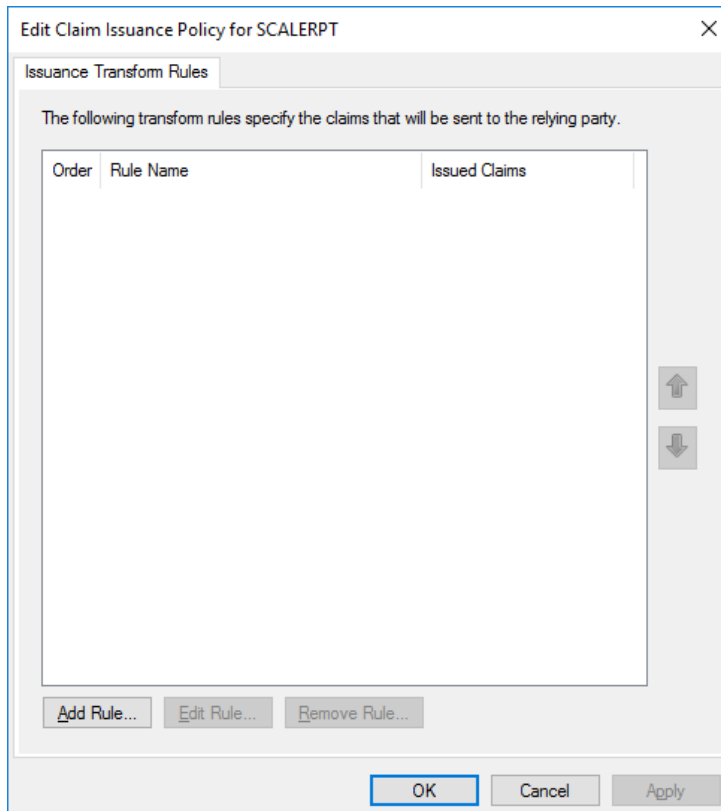
Figure 48



13. Click **Add Rule**.

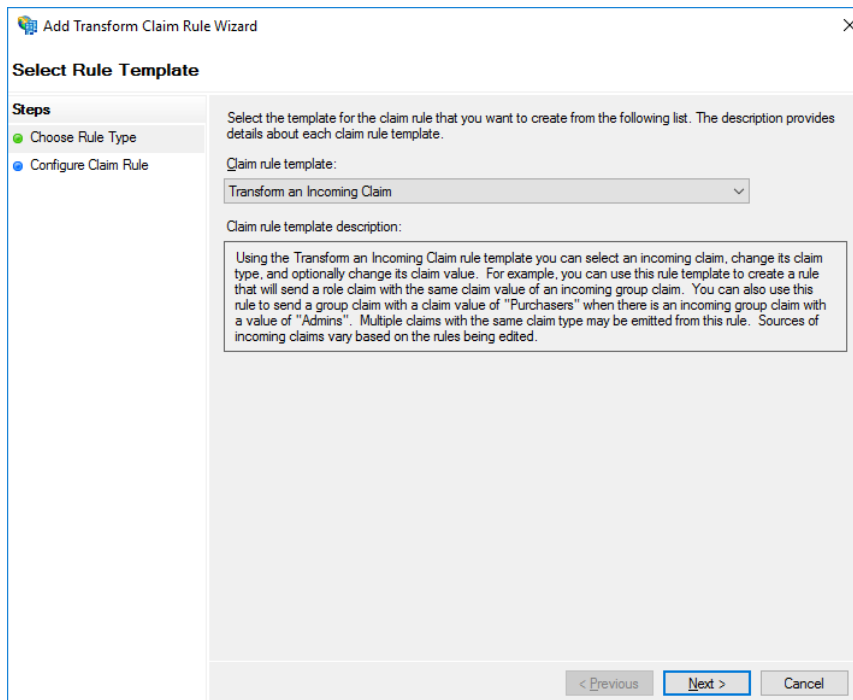
Figure 49

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14. Select **Transform an Incoming Claim** for the **Claim rule template** and click **Next**.

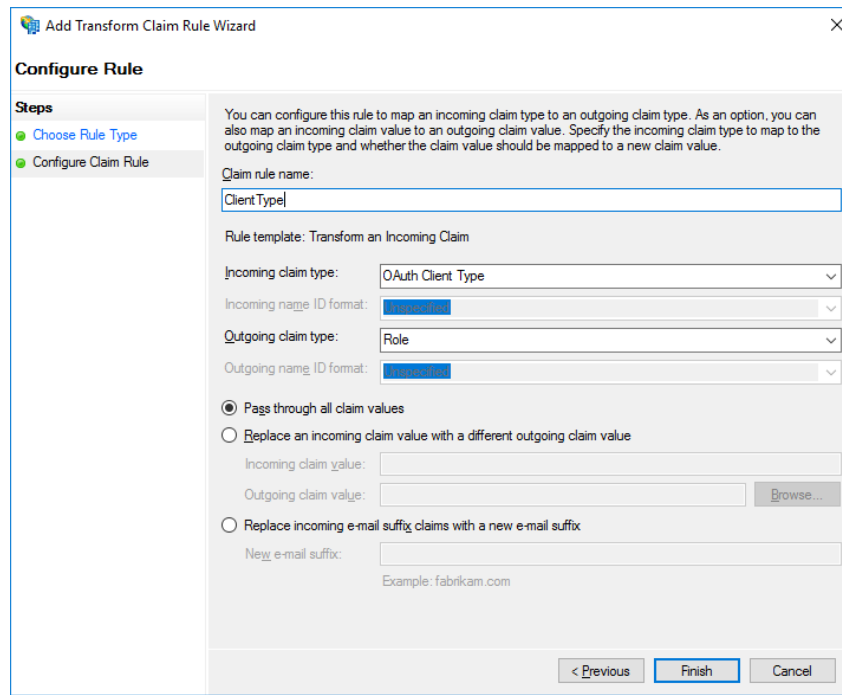
Figure 50



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15. Type the **Claim rule name**, Select **Incoming claim type** as “OAuth Client Type”, **Outgoing claim type** as “Role” and click **Finish**.

Figure 51



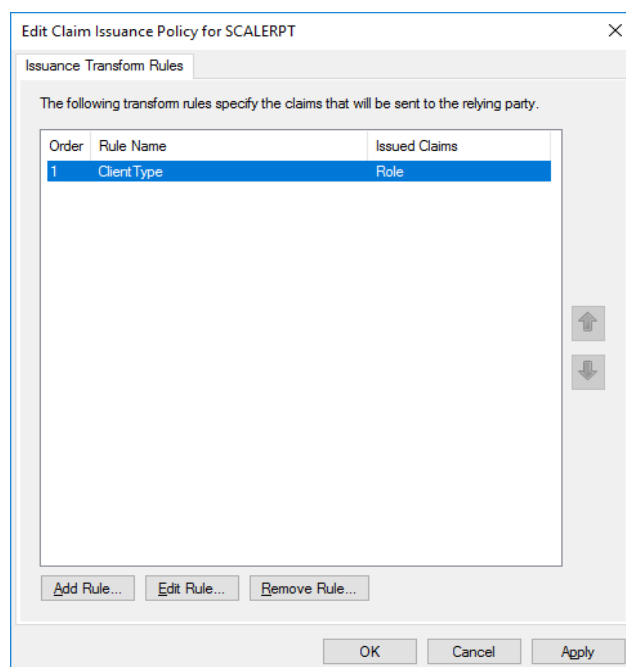
The screenshot shows the 'Add Transform Claim Rule Wizard' window, specifically the 'Configure Rule' step. The 'Steps' pane on the left shows 'Choose Rule Type' and 'Configure Claim Rule' as completed steps. The main area contains the following fields and options:

- Claim rule name:** A text box containing 'ClientType'.
- Rule template:** A dropdown menu set to 'Transform an Incoming Claim'.
- Incoming claim type:** A dropdown menu set to 'OAuth Client Type'.
- Incoming name ID format:** A dropdown menu set to 'urn:oasis:names:tc:oid:1.3.6.1.5.5.2.1.1'.
- Outgoing claim type:** A dropdown menu set to 'Role'.
- Outgoing name ID format:** A dropdown menu set to 'urn:oasis:names:tc:oid:1.3.6.1.5.5.2.1.1'.
- Options:**
 - ☒ Pass through all claim values
 - ☐ Replace an incoming claim value with a different outgoing claim value
 - Incoming claim value:** A text box.
 - Outgoing claim value:** A text box with a 'Browse...' button.
 - ☐ Replace incoming e-mail suffix claims with a new e-mail suffix
 - New e-mail suffix:** A text box with the example 'fabrikam.com'.

At the bottom right are buttons for '< Previous', 'Finish', and 'Cancel'.

16. Click **Apply**.

Figure 52



The screenshot shows the 'Edit Claim Issuance Policy for SCALERPT' window, specifically the 'Issuance Transform Rules' tab. The main area contains a table with the following data:

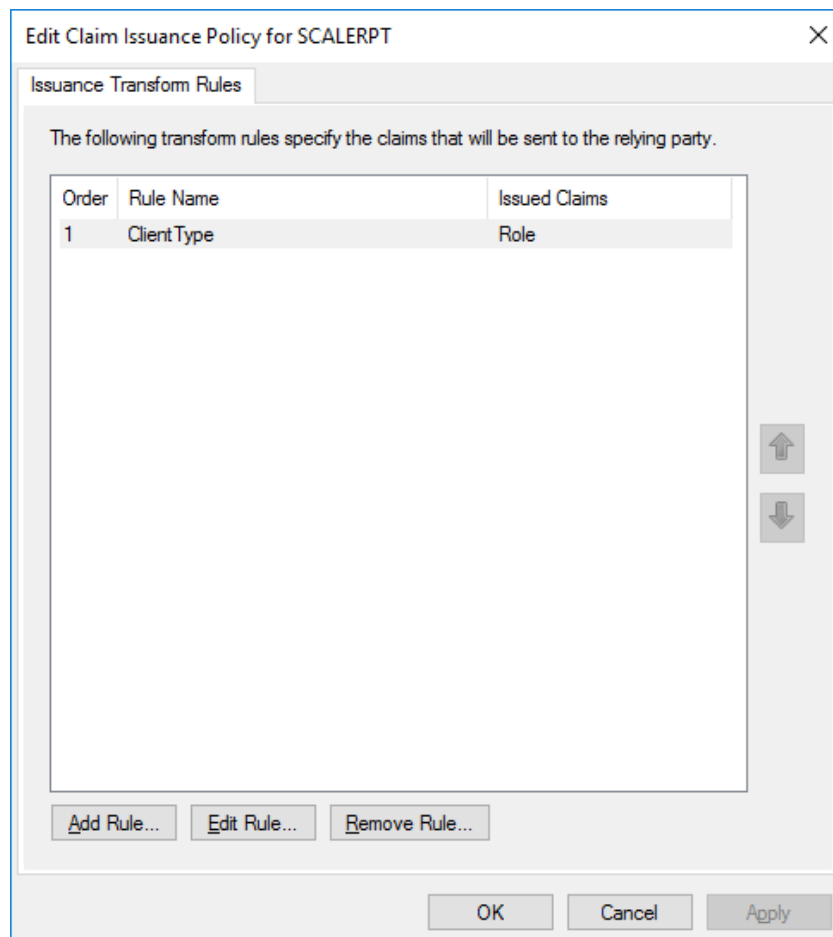
Order	Rule Name	Issued Claims
1	ClientType	Role

Below the table are buttons for 'Add Rule...', 'Edit Rule...', and 'Remove Rule...'. At the bottom right are buttons for 'OK', 'Cancel', and 'Apply'.

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17. Click **OK**.

Figure 53



18. Now open PowerShell as an administrator and run the below command.

```
Grant-AdfsApplicationPermission -ClientRoleIdentifier "<ClientId>" -  
ServerRoleIdentifier "<Relying party Identifier>" -ScopeNames  
"allatclaims", "openid"
```

Change the highlighted <ClientId> to the identifier [copied above](#). Change the highlighted <Relying party Identifier> to Relying party trust identifier [used here](#).

Figure 54

```
PS C:\Windows\system32> Grant-AdfsApplicationPermission -ClientRoleIdentifier "bcefd1b-c9bb-4e3e-a507-4d2cbfa2139c" -Se  
verRoleIdentifier "SCALERPT" -ScopeNames "allatclaims", "openid"  
PS C:\Windows\system32>
```

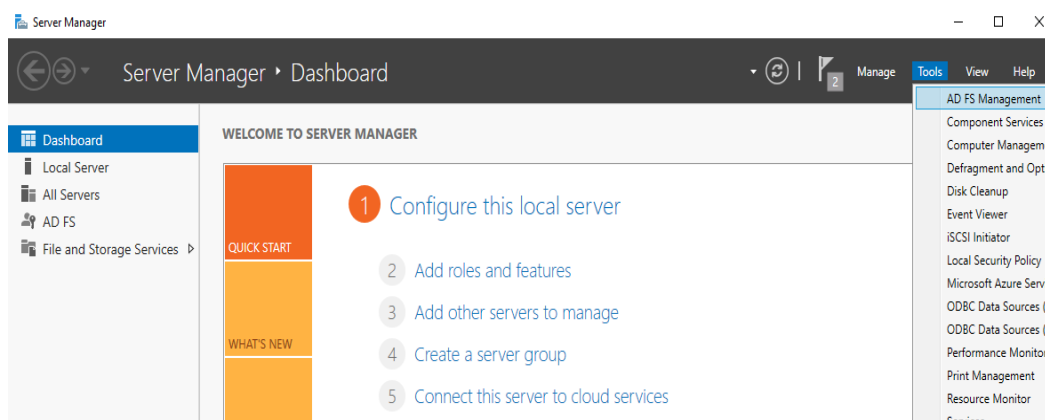
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API Confidential Application [Optional]

This is an optional application only required if custom code like MHE, exit points, Azure functions, host system etc. needs to call SCALE API marked as the SCALEUser role. This application uses OAuth 2.0 client credentials grant flow. The SCALEAPI user needs to be manually created in the SCALE User Profile configurations to make use of this flow. See SDK topic Integration Services > REST-Based Web Services > Authentication in REST-Based Web Services for how to use this application. Below are the steps to create this application.

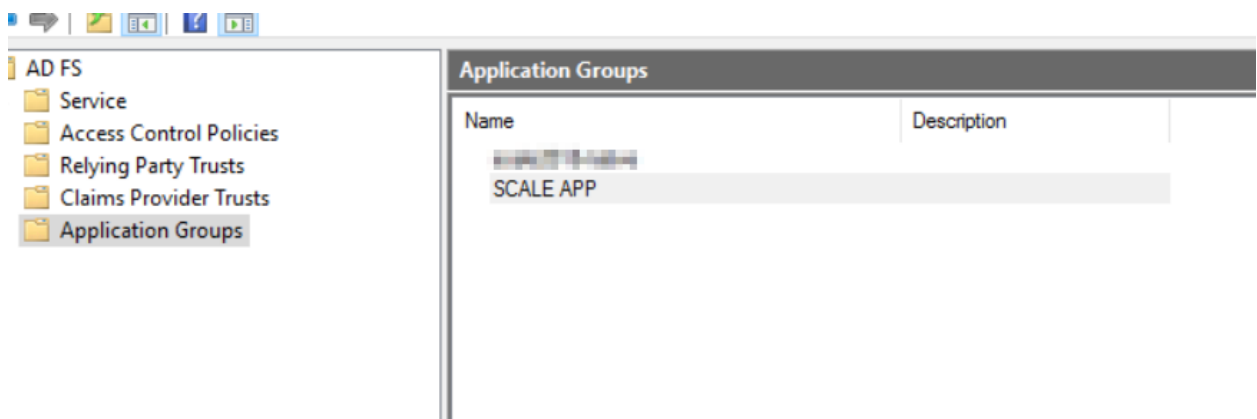
1. Log on to AD FS server and navigate to **Server Manager->Tools->AD FS Management**.

Figure 55



2. Select **Application Groups**.

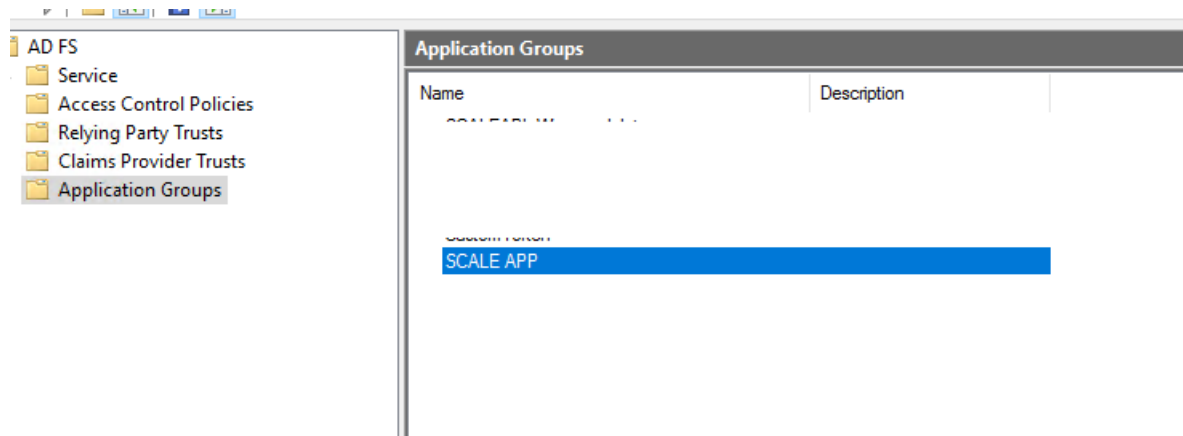
Figure 56



3. From the **Application Groups** window, double click the application created above.

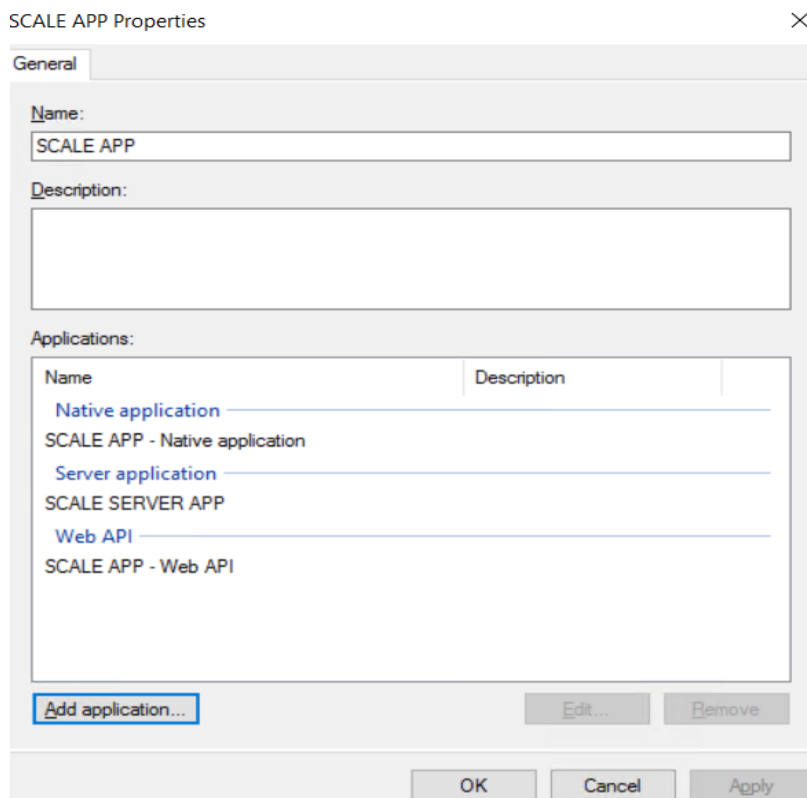
Figure 57

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- Click **Add Application**.

Figure 58



- Select Server application from standalone applications and click **Next**.

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Add a new application to SCALE APP ×

Welcome

Steps

- Welcome
- Server application
- Configure Application Credentials
- Summary
- Complete

Name: SCALE APP

Description:

Template:

Standalone applications

- Native application
- Server application
- Web API

More information...

< Previous Next > Cancel

6. In the server application box, add the below application **Redirect URI** in lowercase. Then click **Add**.

<https://<hostname>/scale/security/postredirecturi>

Note: Hostname should be as discussed above ([Without SCALE environment](#)/ [With SCALE environment](#))

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Add a new application to SCALE APP ×

Server application

Steps

- Welcome
- Server application
- Configure Application Credentials
- Summary
- Complete

Name:
SCALE APP - Server application

Client Identifier:
01f36915-5742-4306-8176-9c6695c42134

Redirect URI:

Description:

< Previous Next > Cancel

7. Click **Next**.

Figure 7

Add a new application to SCALE APP ×

Server application

Steps

- Welcome
- Server application
- Configure Application Credentials
- Summary
- Complete

Name:
SCALE APP - Server application

Client Identifier:
01f36915-5742-4306-8176-9c6695c42134

Redirect URI:

Description:

< Previous **Next >** Cancel

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8. Select **Generate a shared secret**. Click **Copy to clipboard** and save the client secret to later use in a custom application. Click **Next**.

Figure 8

The screenshot shows the 'Add Application Group Wizard' window with the 'Configure Application Credentials' step selected in the left-hand 'Steps' pane. The main area contains the following options:

- ☐ Register a key used to sign JSON Web Tokens for authentication. Below this is a 'Configure...' button.
- ☐ Windows Integrated Authentication. Below this is a 'Select the AD Account:' label, a text box containing 'Example: CONTOSO\expensevc', and a 'Select...' button.
- ☒ Generate a shared secret. Below this is a 'Secret:' label, a text box containing '4r4a1U2nNUlsESZ9VuZ7BAcB4tqB_jvJo_hyKO8L', and a 'Copy to clipboard' button.

A yellow information box at the bottom of the main area contains the text: 'Copy and save the secret. You will not be able to view the secret after the application group is created. You can reset the secret later if required.'

At the bottom of the window are three buttons: '< Previous', 'Next >', and 'Cancel'.

9. Note the **Identifier** for the app. You will configure your application to use this later as API confidential. Click **Next**.

Figure 9

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Add a new application to SCALE APP

Summary

Steps

- Welcome
- Server application
- Configure Application Credentials
- Summary
- Complete

Review the following settings and click 'Next' to create the application.

Application Group

Name: SCALE APP

Server application

Name: SCALE APP - Server application

Identifier: 01f36915-5742-4306-8176-9c6695c42134

Redirect URLs:

<https://hdalal16env1.jlsnetdev.pri/scale/security/postredirecturi>

Use client secret: True

< Previous Next > Cancel

10. Click **Close**.

Figure 10

Add Application Group Wizard

Finish

Steps

- Welcome
- Server application
- Configure Application Credentials
- Summary
- Complete

The Application Group has been successfully created.

Close

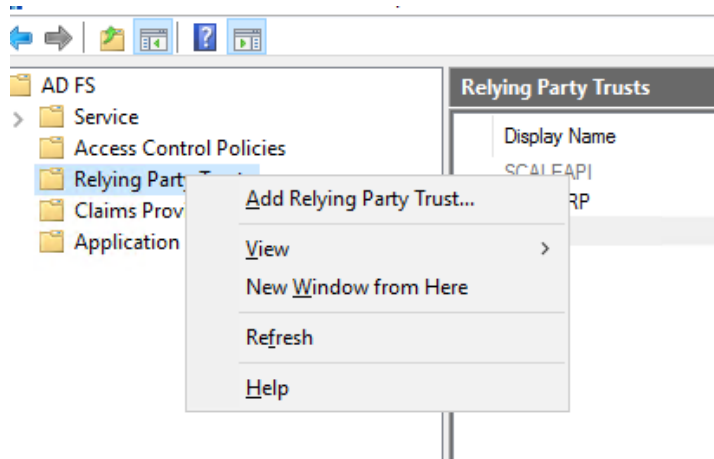
SINGLE SIGN-ON GUIDE: ACTIVE DIRECTORY FEDERATION SERVICES

Configure Relying Party Trust

We need to configure a Relying Party Trust for server-to-server calls to get a custom claim for the application.

1. From AD FS explorer, right-click the **Relying Party Trust** folder and click **Add Relying Party Trust**.

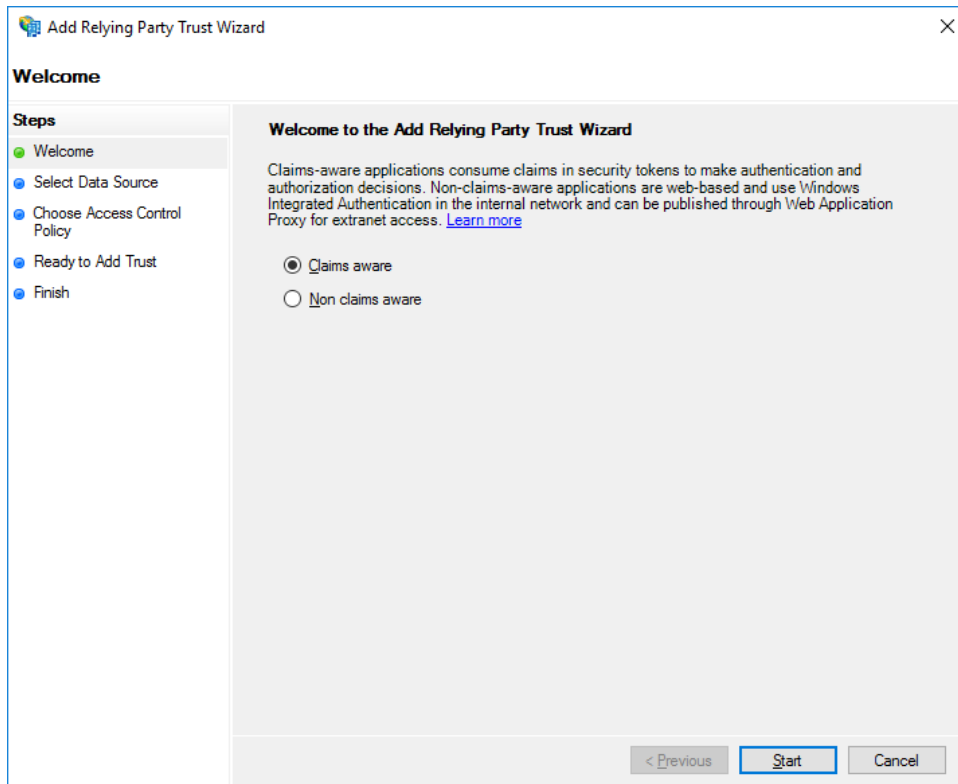
Figure 36



2. Select **Claims aware** and click **Start**.

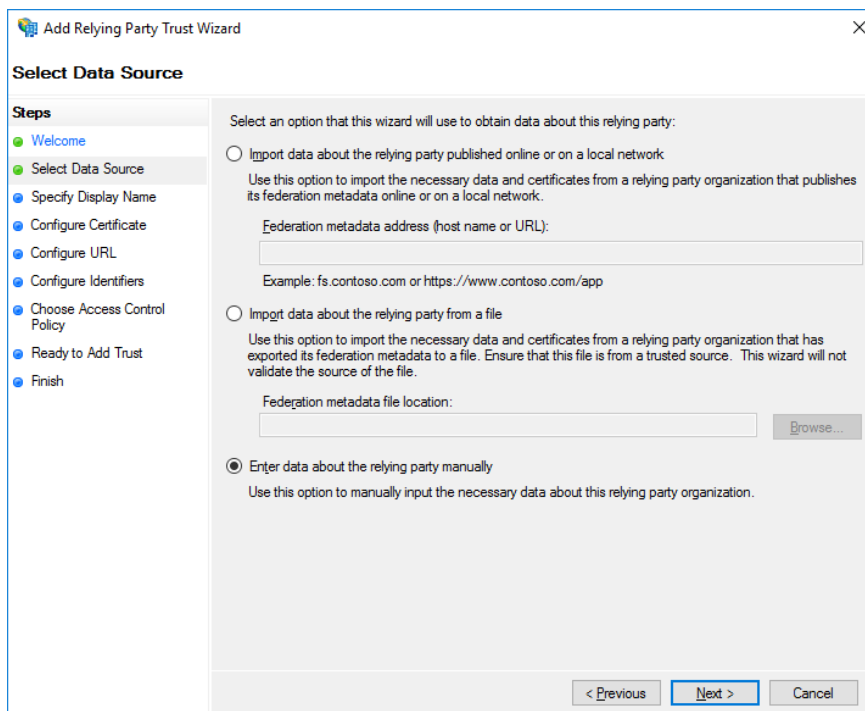
Figure 37

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3. Select **Enter data about the relying party manually** and click **Next**.

Figure 38



4. Enter the **Display name** and click **Next**.

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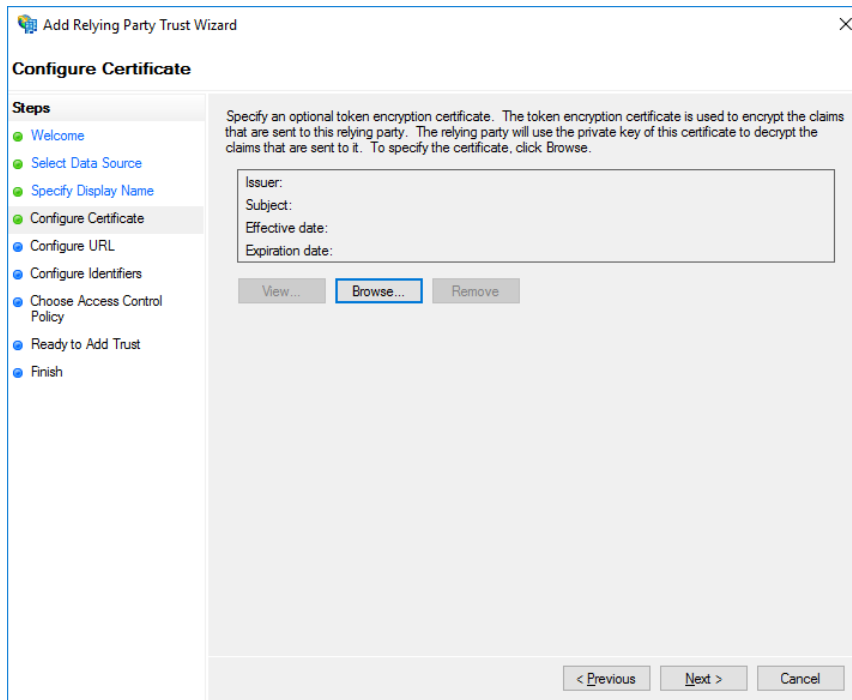
Figure 39

The screenshot shows the 'Add Relying Party Trust Wizard' window. The title bar reads 'Add Relying Party Trust Wizard' with a close button. The window is titled 'Specify Display Name'. On the left, a 'Steps' pane lists the following steps: Welcome, Select Data Source, Specify Display Name (current step), Configure Certificate, Configure URL, Configure Identifiers, Choose Access Control Policy, Ready to Add Trust, and Finish. The main area contains the instruction 'Enter the display name and any optional notes for this relying party.' Below this, there is a 'Display name:' label followed by a text box containing 'SCALEAPI'. Below the text box is a 'Notes:' label followed by a large text area. At the bottom right, there are three buttons: '< Previous', 'Next >' (highlighted with a blue border), and 'Cancel'.

5. Click **Next**.

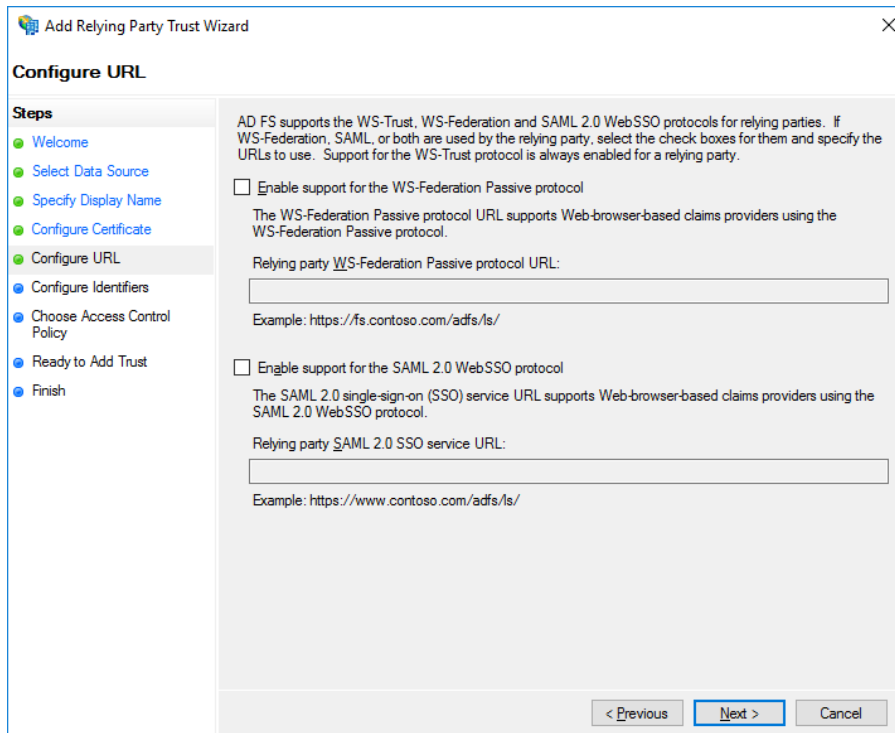
Figure 40

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6. Click **Next**.

Figure 41



7. Enter **Relying party trust identifier** as an identifier of the server application created [above](#).

Figure 43

SINGLE SIGN-ON GUIDE: ACTIVE DIRECTORY FEDERATION SERVICES

Add Relying Party Trust Wizard

Configure Identifiers

Steps

- Welcome
- Select Data Source
- Specify Display Name
- Configure Certificate
- Configure URL
- Configure Identifiers**
- Choose Access Control Policy
- Ready to Add Trust
- Finish

Relying parties may be identified by one or more unique identifier strings. Specify the identifiers for this relying party trust.

Relying party trust identifier:

01f36915-5742-4306-8176-9c6695c42134 Add

Example: <https://fs.contoso.com/adfs/services/trust>

Relying party trust identifiers:

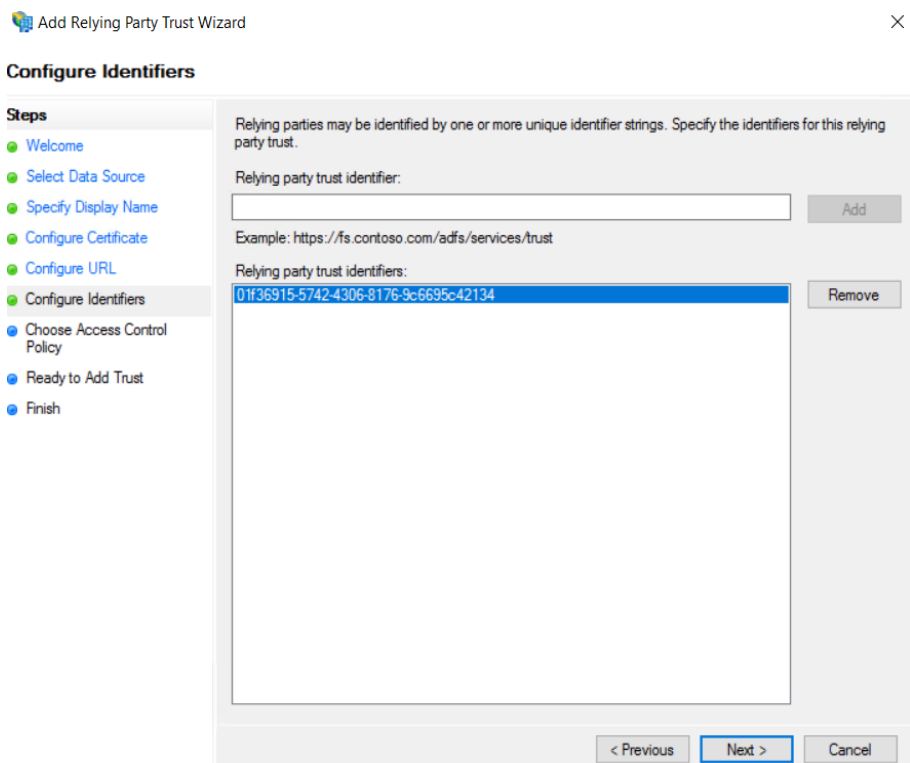
Remove

< Previous Next > Cancel

- Click **Add** and then click **Next**.

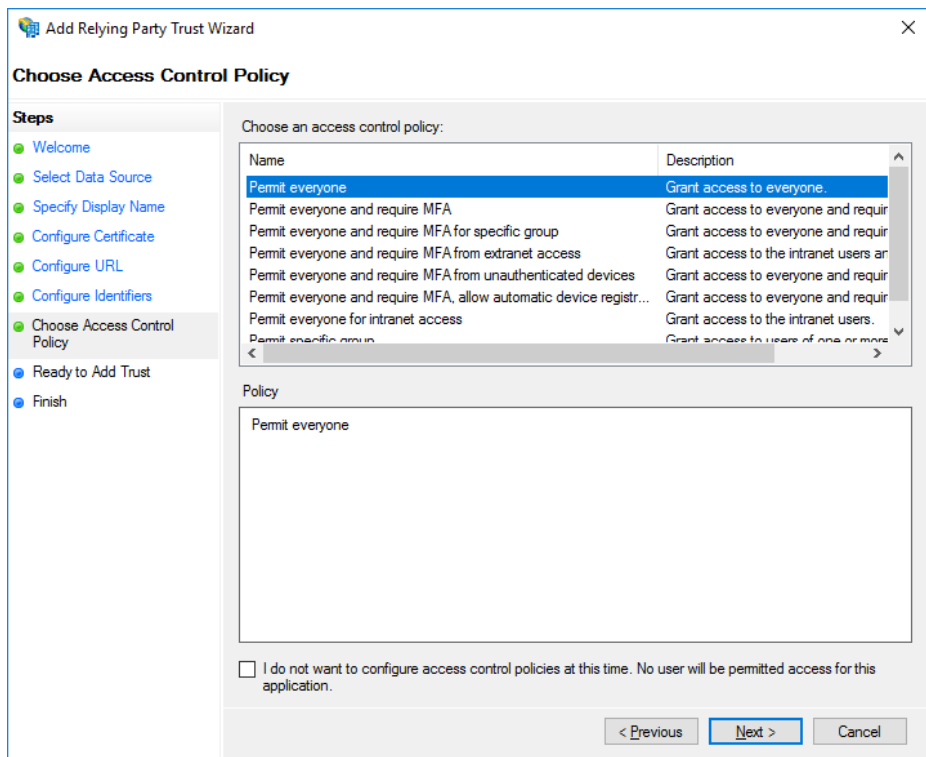
Figure 44

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9. Click **Next**.

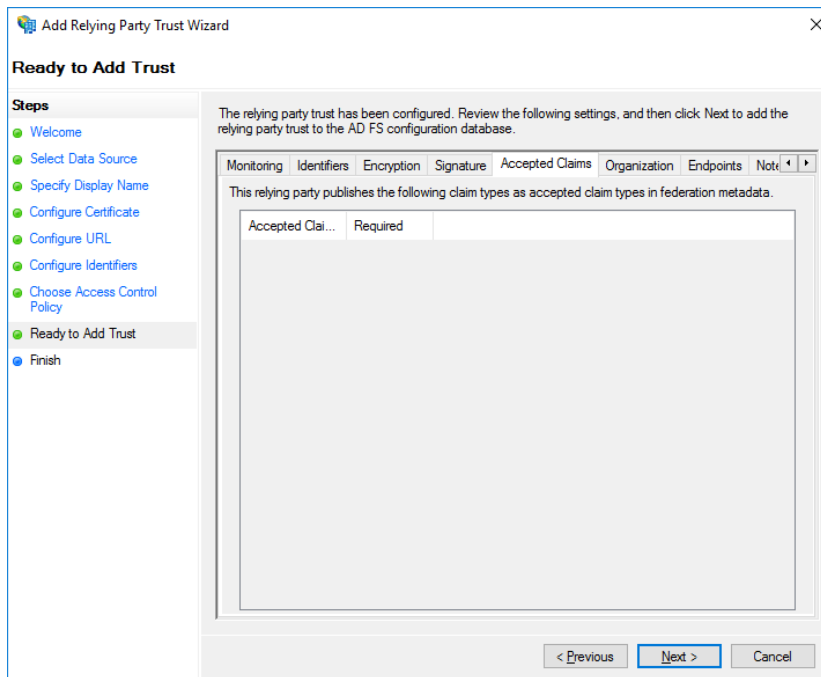
Figure 45



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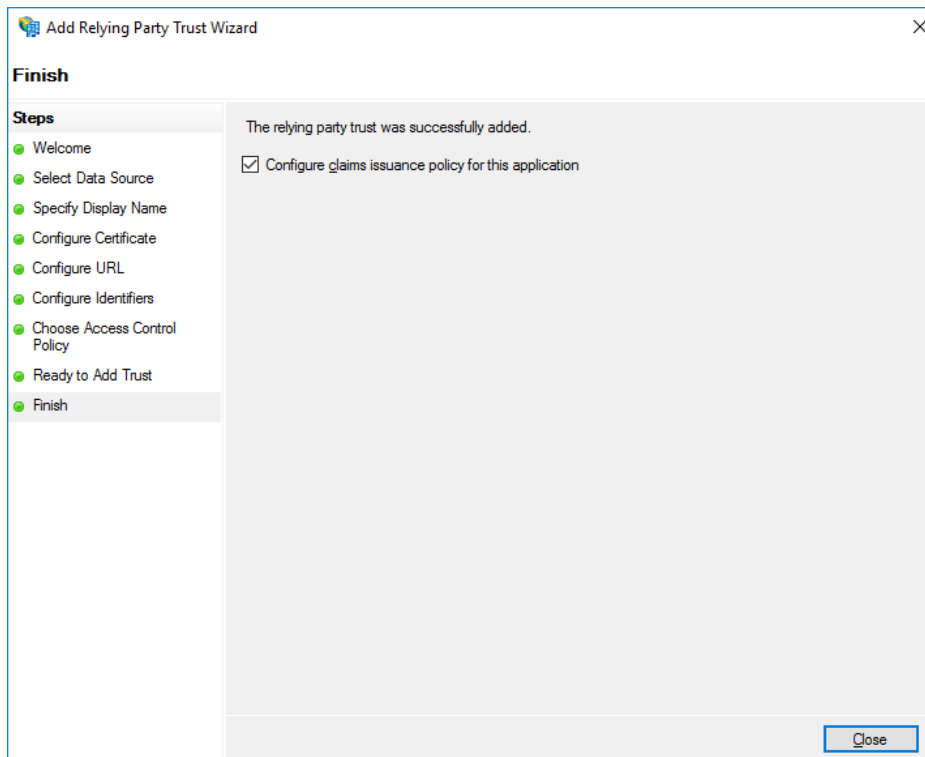
10. Click **Next**.

Figure 46



11. Click **Close**.

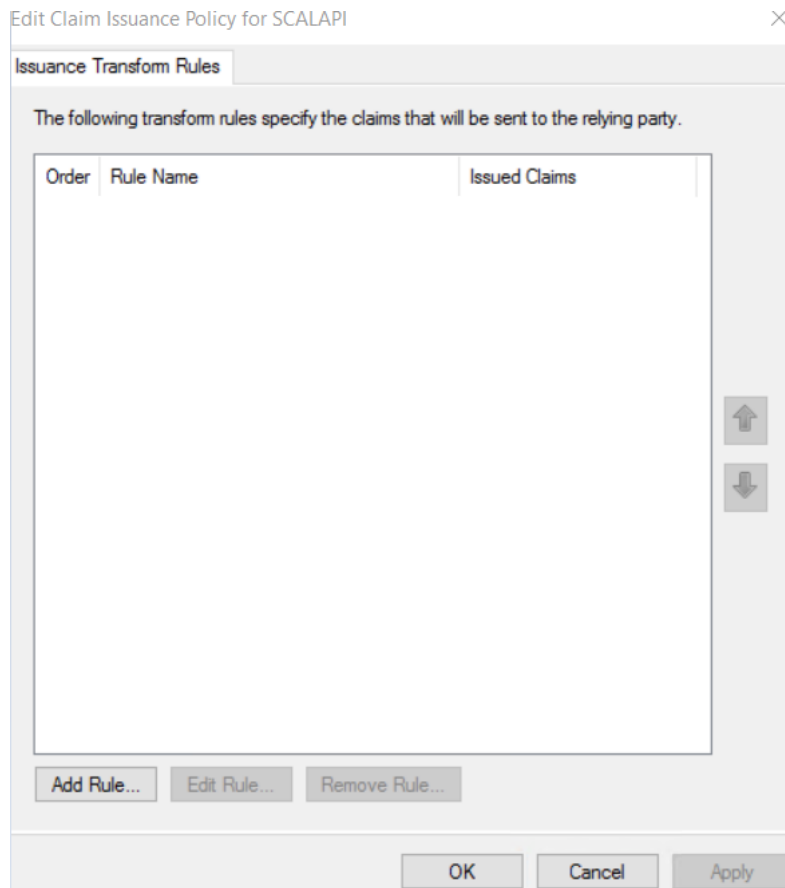
Figure 47



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12. Click **Add Rule**.

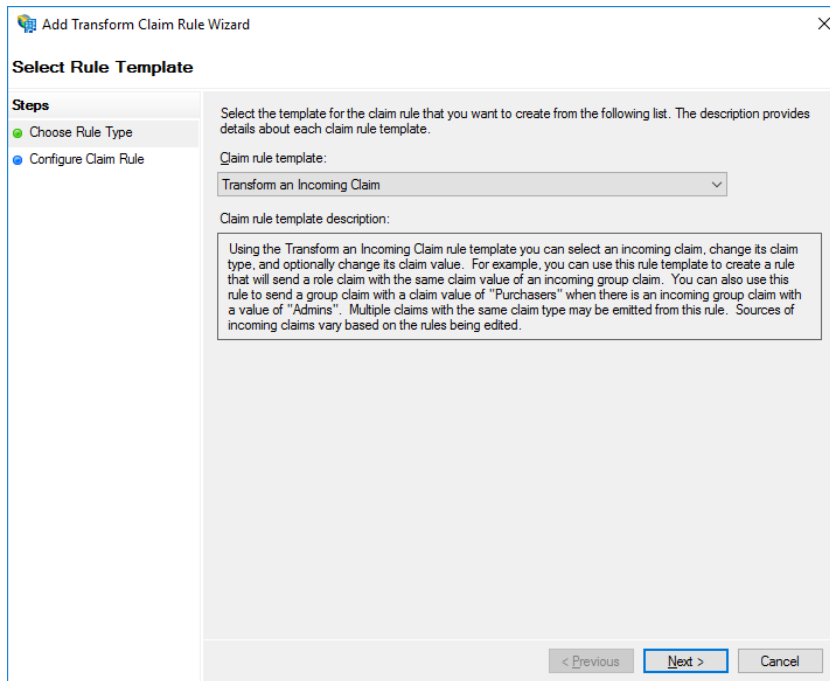
Figure 49



13. Select **Transform an Incoming Claim** for the **Claim rule template** and click **Next**.

Figure 50

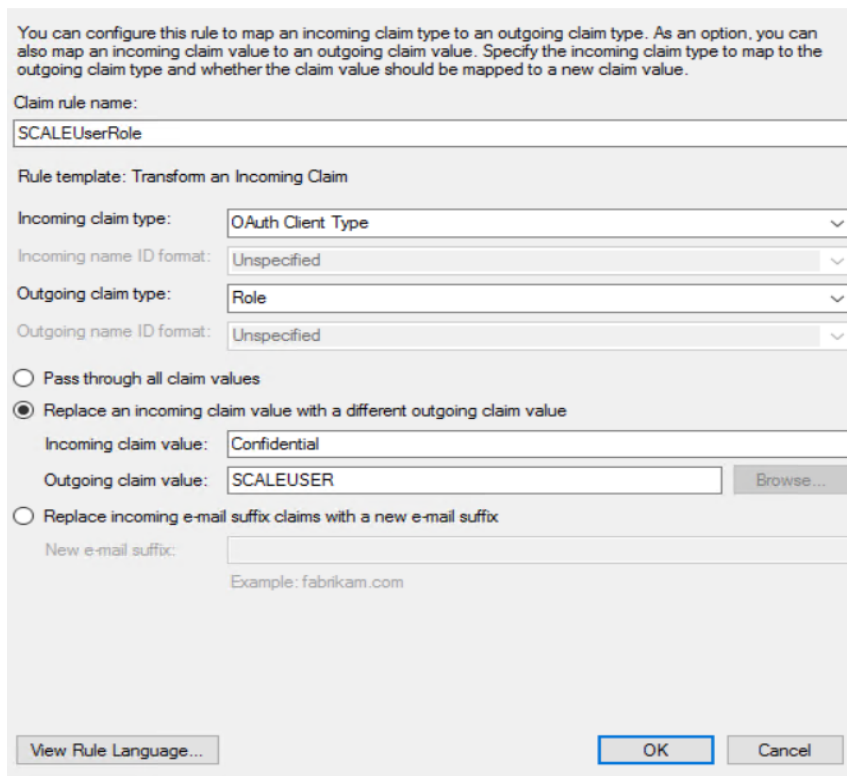
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The screenshot shows the 'Add Transform Claim Rule Wizard' window, specifically the 'Select Rule Template' step. On the left, a 'Steps' pane shows 'Choose Rule Type' as the current step and 'Configure Claim Rule' as the next. The main area contains instructions to select a template, a dropdown menu for 'Claim rule template' set to 'Transform an Incoming Claim', and a text box with a detailed description of this template. At the bottom, there are navigation buttons: '< Previous', 'Next >', and 'Cancel'.

14. Type the **Claim rule name**, Select **Incoming claim type** as “OAuth Client Type”, **Outgoing claim type** as “Role”, modify values as below, and click **Finish**.

Figure 51

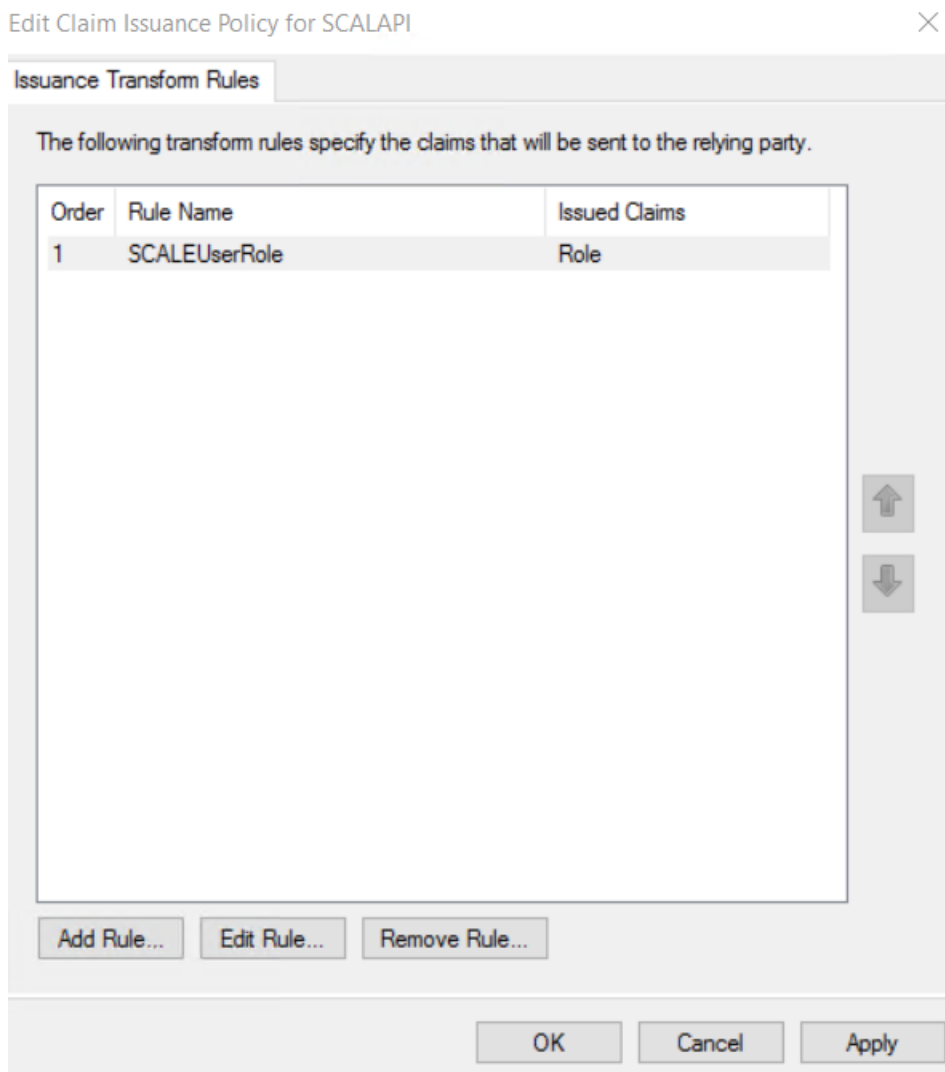


The screenshot shows the configuration window for a 'Transform an Incoming Claim' rule. It includes fields for 'Claim rule name' (SCALEUserRole) and 'Rule template' (Transform an Incoming Claim). Below these are dropdowns for 'Incoming claim type' (OAuth Client Type), 'Incoming name ID format' (Unspecified), 'Outgoing claim type' (Role), and 'Outgoing name ID format' (Unspecified). There are three radio button options: 'Pass through all claim values', 'Replace an incoming claim value with a different outgoing claim value' (which is selected), and 'Replace incoming e-mail suffix claims with a new e-mail suffix'. The selected option has input fields for 'Incoming claim value' (Confidential) and 'Outgoing claim value' (SCALEUSER), with a 'Browse...' button next to the latter. The 'New e-mail suffix' field is empty, with an example 'fabrikam.com' shown below it. At the bottom, there are buttons for 'View Rule Language...', 'OK', and 'Cancel'.

15. Click **Apply** and **OK**.

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Figure 52



16. Now open PowerShell as an administrator and run the below command.

```
Grant-AdfsApplicationPermission -ClientRoleIdentifier "<Identifier>"  
-ServerRoleIdentifier "<Identifier>" -ScopeNames  
"allatclaims","openid"
```

Change the highlighted <Identifier> to the Identifier [copied from above](#)

Figure 54



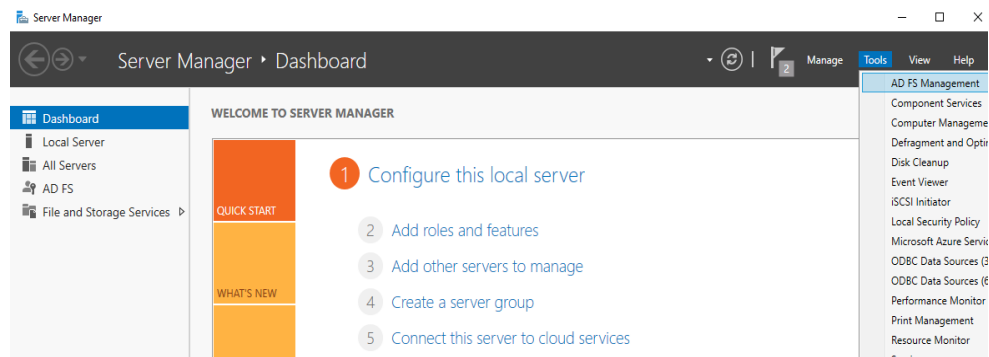
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Register SCALE as Native Application

For the SCALE Desktop to work with AD FS, we need to register SCALE as a Native application in AD FS.

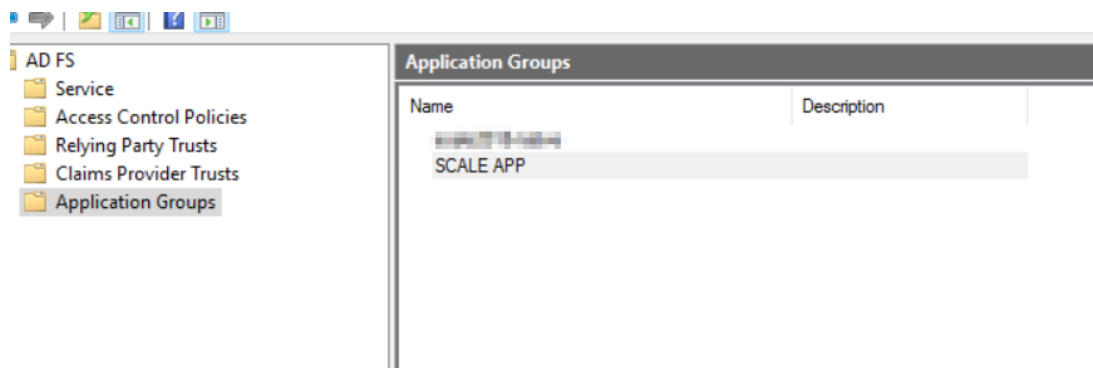
11. Log on to AD FS server and navigate to **Server Manager->Tools->AD FS Management**.

Figure 55



12. Select **Application Groups**.

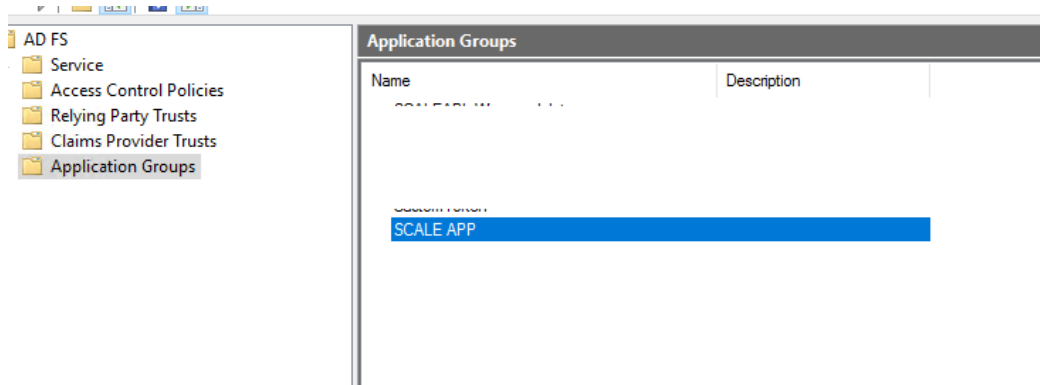
Figure 56



13. From the **Application Groups** window, double click the application created above.

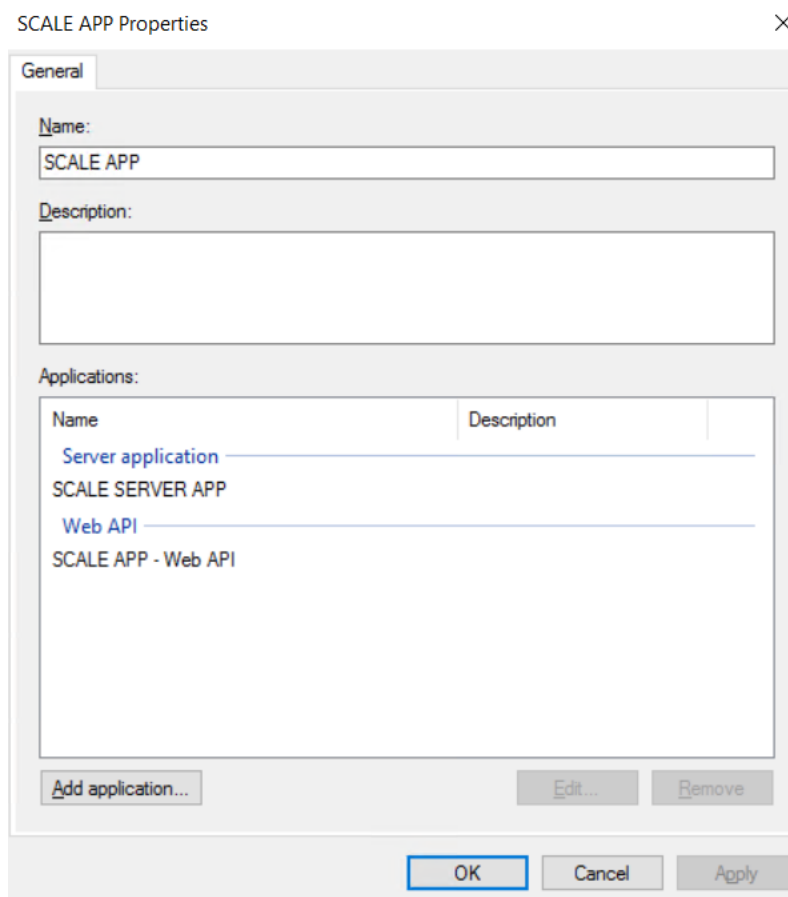
Figure 57

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14. Click **Add Application**.

Figure 58



15. Select Native application from standalone applications and click **Next**.

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16. In the Native application box, add the below application **Redirect URIs** in lower case. Then click **Add**.

<https://<hostname>/scale/security/postredirecturi>

<https://<hostname>/scale/security/warehousemobilepostredirecturi>

Note: Hostname should be as discussed above ([Without SCALE environment](#)/ [With SCALE environment](#))

Figure 59

Add a new application to SCALE APP

Native application

Steps

- Welcome
- Native application
- Summary
- Complete

Name: SCALE APP - Native application

Client Identifier: b02d0f19-10b9-4145-9c70-f4da183fc181

Redirect URI: <https://hdalalenv12016.ilnetdev.pri/scale/security/postredirecturi> Add Remove

Description:

< Previous Next > Cancel

17. Click **Next**.

Figure 60

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The screenshot shows a web-based configuration window titled "Add a new application to SCALE APP". The window has a close button (X) in the top right corner. On the left, there is a "Steps" sidebar with four items: "Welcome" (green dot), "Native application" (green dot and highlighted), "Summary" (blue dot), and "Complete" (blue dot). The main area is titled "Native application" and contains the following fields and controls:

- Name:** A text box containing "SCALE APP - Native application".
- Client Identifier:** A text box containing "b02d0f19-10b9-4145-9c70f4da183fc181".
- Redirect URI:** A section with an example text "Example: https://Contoso.com", an "Add" button, a text box containing "https://hda1alenv12016.ilnetdev.pr/scale/security/postredirecturi" (which is highlighted in blue), and a "Remove" button.
- Description:** An empty text box.

At the bottom of the window are three buttons: "< Previous", "Next >" (which is highlighted with a blue border), and "Cancel".

18. Note the **Identifier** for the app. You will configure your application to use this later in SCALE configuration as Native client id. Click **Next**.

Figure 61

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Add a new application to SCALE APP ×

Summary

Steps

- Welcome
- Native application
- Summary
- Complete

Review the following settings and click 'Next' to create the application.

Application Group

Name: SCALE APP

Native application

Name: SCALE APP - Native application

Identifier: b02d0f19-10b9-4145-9c70-f4da183fc181

Redirect URLs:

<https://hdalalenv12016.ilisnetdev.pri/scale/security/postredirecturi>

< Previous **Next >** Cancel

19. Click **Close**.

Figure 62

Add Application Group Wizard ×

Finish

Steps

- Welcome
- Native application
- Summary
- Complete

The Application Group has been successfully created.

Close

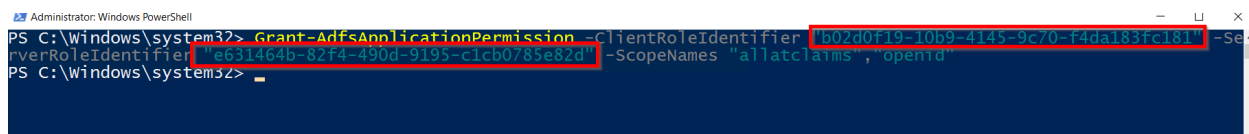
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20. Now open PowerShell as an administrator and run the below command.

```
Grant-AdfsApplicationPermission -ClientRoleIdentifier  
"<NativeClientId>" -ServerRoleIdentifier "<ClientId>" -ScopeNames  
"allatclaims","openid"
```

Change the highlighted <NativeClientId> to the identifier [copied above](#). Change the highlighted <ClientId> to the relying party trust identifier in Figure 61.

Figure 63

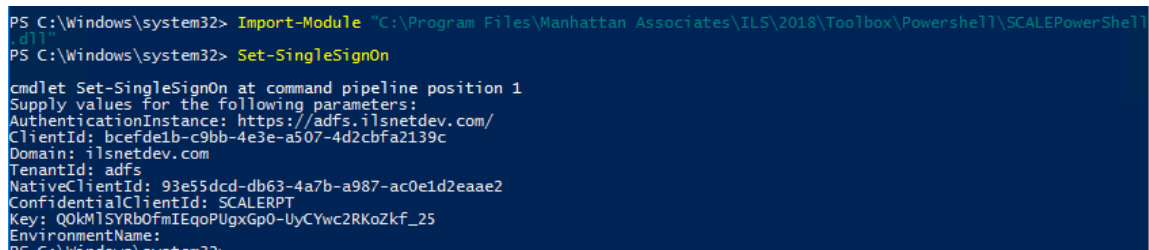
A screenshot of a Windows PowerShell window running as Administrator. The command prompt shows the execution of the 'Grant-AdfsApplicationPermission' command. The parameters are: -ClientRoleIdentifier 'b02d0f19-10b9-4145-9c70-f4da183fc181', -ServerRoleIdentifier 'e631464b-82f4-490d-9195-c1cb0785e82d', and -ScopeNames 'allatclaims', 'openid'. The ClientId and NativeClientId values are highlighted with red boxes.

Configure SCALE to use AD FS

For SCALE to use AD FS, we can configure SSO using the SCALE PowerShell module.

1. Open PowerShell on the SCALE application server as administrator.
2. Run the command `Set-SingleSignOn`
3. Provide the values as follows
 - **AuthenticationInstance:** `https://Federation service Name` [Copied here](#).
 - **ClientId:** [Client Id copied here](#)
 - **Domain:** Domain of AD FS
 - **TenantId:** adfs
 - **NativeClientId:** [Native Client Id copied here](#)
 - **ConfidentialClientId:** Relying party trust identifier [copied here](#)
 - **Key:** secret [copied here](#)
 - **APICConfidentialClientId:** identifier [copied here](#)
 - **EnvironmentName:** This is the name of environment you want to update this SSO value of. Empty string means Default Database

Figure 64

A screenshot of a Windows PowerShell window running as Administrator. The command prompt shows the execution of the 'Set-SingleSignOn' command. The output displays the command pipeline position and the values for the following parameters: AuthenticationInstance: https://adfs.ilsnetdev.com/, ClientId: bcefde1b-c9bb-4e3e-a507-4d2cbfa2139c, Domain: ilsnetdev.com, TenantId: adfs, NativeClientId: 93e55dcd-db63-4a7b-a987-ac0e1d2eaae2, ConfidentialClientId: SCALERPT, Key: Q0kM1SYRb0fmIEqoPUgxGp0-UyCYwc2RKoZkf_25, EnvironmentName: . The command prompt shows the execution of the 'Set-SingleSignOn' command. The output displays the command pipeline position and the values for the following parameters: AuthenticationInstance: https://adfs.ilsnetdev.com/, ClientId: bcefde1b-c9bb-4e3e-a507-4d2cbfa2139c, Domain: ilsnetdev.com, TenantId: adfs, NativeClientId: 93e55dcd-db63-4a7b-a987-ac0e1d2eaae2, ConfidentialClientId: SCALERPT, Key: Q0kM1SYRb0fmIEqoPUgxGp0-UyCYwc2RKoZkf_25, EnvironmentName: .

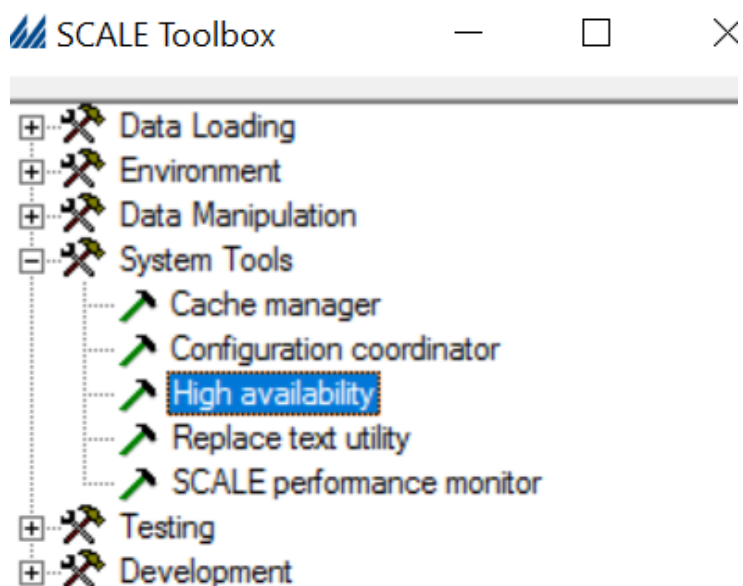
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Note: Once we update the SSO values it is mandatory to do an ILS Reset (proceed to C:\Program Files\Manhattan Associates\ILS\<version>\Utilities and then right click on ILS Reset Services and run as administrator) for the values to be reflected.

Configure load balancer

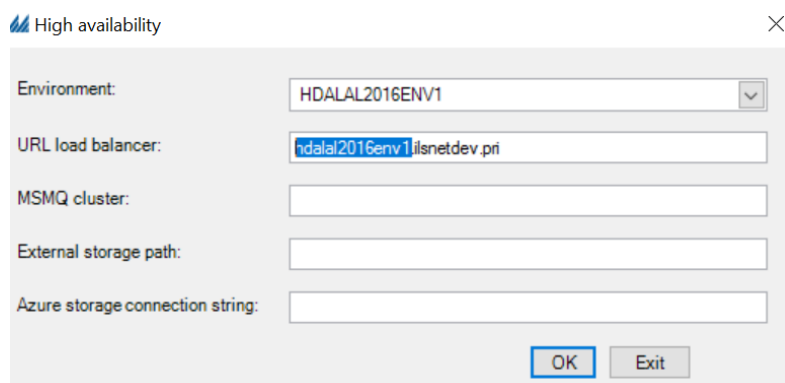
1. Open SCALE toolbox as administrator.
2. Navigate to System Tools and double click High availability option

Figure 65



3. Update the URL load balancer as discussed above ([Without SCALE environment/](#) [With SCALE environment](#))

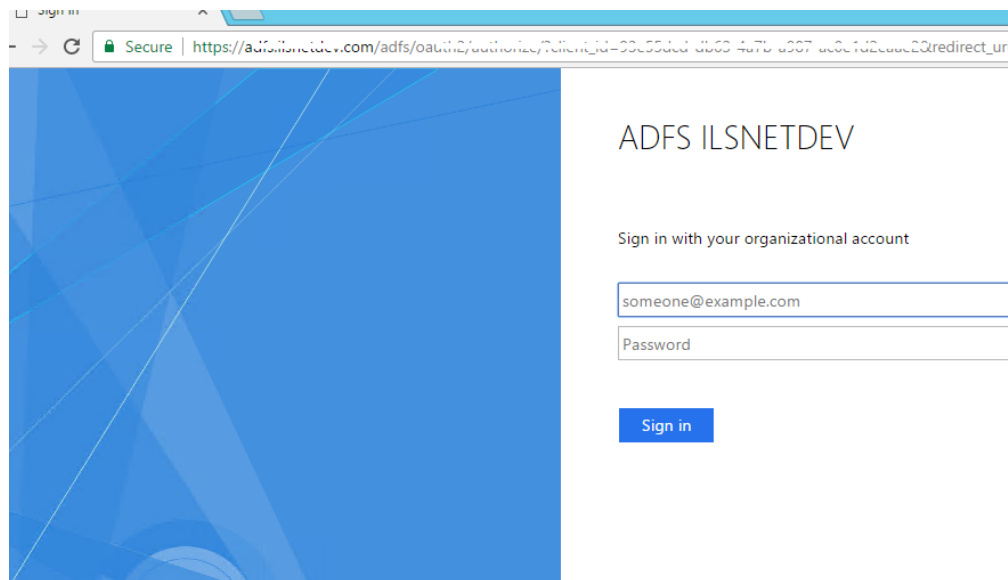
Figure 66



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Now open a browser and navigate to the SCALE application using the full domain name (i.e. <https://hostname/SCALE/security/login>). You will be redirected to your AD FS server <https://adfs.ilsnetdev.com> and asked for your domain login id and password if you are not logged in with your domain account already.

Figure 67



Appendix 1

Customer Upgrading from 2021/2022 to 2023

If you are upgrading from 2021/2022 to 2023, please follow steps 15 to 20 in [Configure Custom Claims](#) section.

Customizing the Login Page

Customizing the login page for Microsoft Entra ID and AD FS (with company logo, etc.) is outlined in the SDK See the **User Interface Extensibility > Metadata Web Pages > Single Signon Page > How to: Customize the Single Signon Page** topic for more information.

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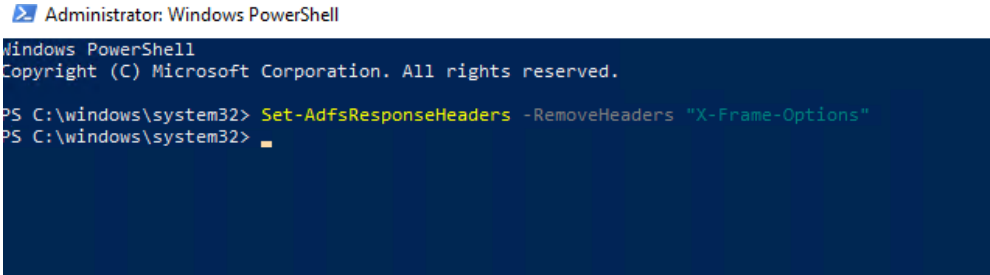
Troubleshooting

- Use SSL certificates from a public trusted authority for AD FS and SCALE. If using self-signed certificates, ensure that the root certificate is trusted on all application servers, AD FS server, and all browser clients using SCALE web app.
- Use the same hostname to log into the SCALE web application as mentioned in redirect URL in AD FS.
- AD FS redirect URL is case sensitive. Use the exact same URL in web config as provided in AD FS.
- On ADFS 5(AD FS on windows server 2019) Microsoft added new functionality to restrict silent token retrieval in iframe by passing new response header X-Frame-Options to deny which limit SCALE web/mobile applications to get token silently. For more info see <https://docs.microsoft.com/en-us/windows-server/identity/ad-fs/operations/customize-http-security-headers-ad-fs>

There is a workaround which will remove default header X-Frame-Options.

- a. Run following PowerShell command of AD FS server

Set-AdfsResponseHeaders -RemoveHeaders "X-Frame-Options"



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\windows\system32> Set-AdfsResponseHeaders -RemoveHeaders "X-Frame-Options"
PS C:\windows\system32>
```

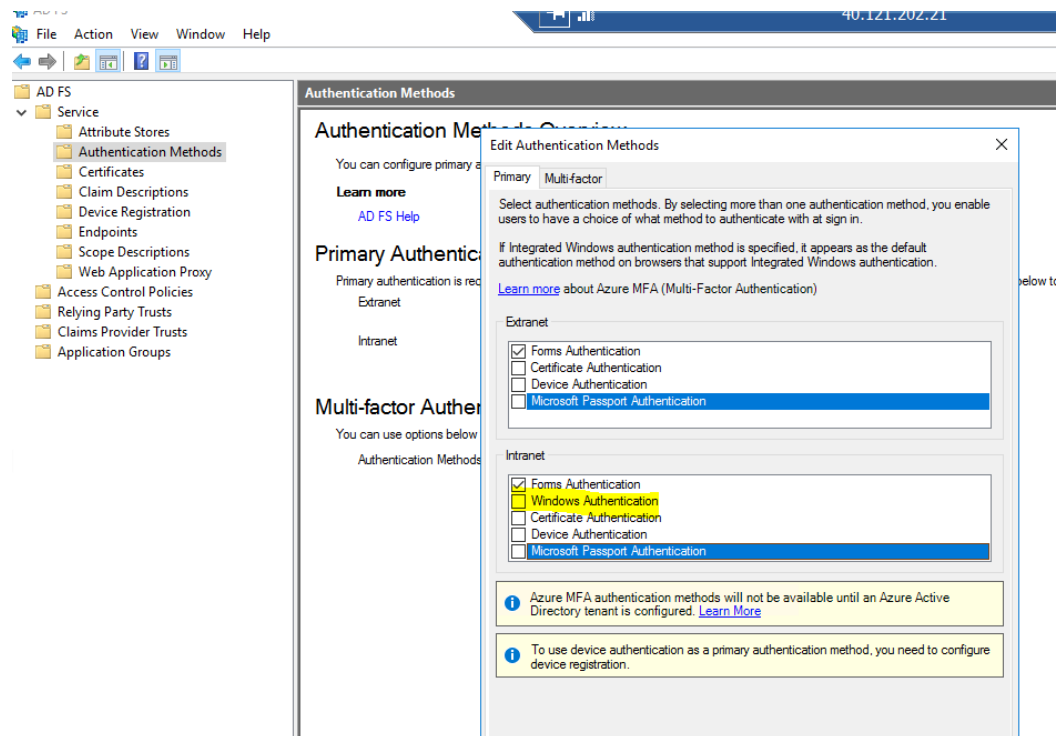
Not supported

AD FS direct/indirect (Microsoft Entra redirect to AD FS) Windows Integrated Authentication (WIA) is not supported for Android RF SCALE/Warehouse Mobile application as Android web views do not support Windows Integrated Authentication. There are a few workarounds.

1. Disable windows authentication for intranet Zone. This will affect all AD FS applications on all agents.

Figure 68

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2. Configuring intranet forms-based authentication for devices that do not support WIA. See <https://docs.microsoft.com/en-us/windows-server/identity/ad-fs/operations/configure-intranet-forms-based-authentication-for-devices-that-do-not-support-wia>. This will affect all AD FS applications but only on selected agents.